Online Supplement

A Sufficient Statistic for Algorithmic Feedback

Niroula, Niroula and Khatiwada, core version, 12 November 2026

The supplement contains the proofs of the results stated in the main text (Section S1), the further equilibrium results with their proofs (Section S2), execution cost and capital (Section S3), the two-group case used by Theorem 5.4(ii) (Section S4), the supervisory decision in detail (Section S5), the proofs for saturating traders (Section S6), machine checks in detail (Section S7), the machine-checked declarations (Section S8) and the notation (Section S9). Proofs are taken unchanged from the full working paper of 5 November 2026. Only cross-references to material not included here have been changed.

S1 Proofs of the main results

A lemma used in the proofs.

Lemma S1.1 (Positive price-response geometry)**.** *In the positive, stable economy the impulse response is a nonnegative mixture of decaying exponentials,*

$$h_j = \sum_\ell \omega_\ell \lambda_\ell^j, \qquad \omega_\ell \geq 0, \quad \sum_\ell \omega_\ell = \kappa, \quad 0 \leq \lambda_\ell < 1.$$

It is completely monotone and nonincreasing; its peak is the direct impact κ, even though J is non-normal. In price/memory coordinates the price-only gain is $\sup_t \|e_x^\top J^t\|_2 = 1$.

Lemma S1.1. Omit zero-weight cohorts. For $a > 0$ put $b_i = \kappa q_i$ and

$$T = \operatorname{diag}(\sqrt{a}, \sqrt{b_1/\eta_1}, \ldots), \qquad u = (\sqrt{a}, \sqrt{b_1\eta_1}, \ldots)^\top.$$

The price-first transition becomes the symmetric matrix $J' = TJT^{-1} = \operatorname{diag}(0, r_1, \ldots) + uu^\top$, the input transforms to κu and $J'e_x = \sqrt{a}\,u$, so $h_j = (\kappa/a)e_x^\top (J')^{j+1} e_x$. The spectral decomposition of J' gives nonnegative weights summing to $h_0 = \kappa$, and Schur stability puts the eigenvalues in $[0, 1)$. The case $a = 0$ follows by continuity. A nonnegative mixture of decaying geometric sequences is completely monotone. Since J is nonnegative with row sums at most one, every price row of J^t has ℓ_1, hence ℓ_2, norm at most one, attained at $t = 0$.

Lemma 2.1. (i) For $v_{t+1} = v_t + \zeta_{t+1}$ and $r_t = v_t + \nu_t$, the steady-state prior variance P solves $P = P\sigma_\nu^2/(P+\sigma_\nu^2) + \sigma_\zeta^2$, i.e. $P^2 - \sigma_\zeta^2 P - \sigma_\zeta^2\sigma_\nu^2 = 0$, whose positive root is $P = (\sigma_\zeta^2 + \sqrt{\sigma_\zeta^4 + 4\sigma_\zeta^2\sigma_\nu^2})/2$. The gain is $\eta = P/(P+\sigma_\nu^2) \in (0, 1)$, which simplifies to (7), and the update $\hat{v}_{t+1} = \hat{v}_t + \eta(r_t - \hat{v}_t)$ has the stated form. (ii) $\nabla_\theta \frac{1}{2} H(x_t - \theta)^2 = -H(x_t - \theta)$, so $\theta_{t+1} = (1-\gamma)\theta_t + \alpha H(x_t - \theta_t) = (1-\eta)\theta_t + \eta q x_t$ with $\eta = \gamma + \alpha H$ and $q = \alpha H/\eta$.

Theorem 3.3. With zero initial states, $y_{i,t} = \sum_{s \leq t} \eta_i r_i^{t-s} x_s$, so $\sum_i q_i y_{i,t} = \sum_{s \leq t} k_{t-s} x_s$ and every response is determined by (a, κ, k), hence by M. Conversely, equal responses for all probes imply equal H, hence equal G, and partial fractions of G identify the distinct poles with nonzero residues w_i, i.e. M. After cancellation, $H = \kappa z D(z)/[(z-a)D(z) - \kappa N(z)]$ with $D = \prod(z - r_i)$ and $N(z) = \sum_i w_i z \prod_{j \neq i}(z - r_j)$, as in (4); at a retained r_i the denominator equals $-\kappa N(r_i) =$

$-\kappa w_i r_i \prod_{j\neq i}(r_i - r_j) \neq 0$ (memory rates are positive because $\eta_i < 1$), so no root of D cancels. The numerator $\kappa z D(z)$ has no roots other than 0 and the r_i, so these are the only possible cancellations. The remaining numerator factor z cancels if and only if the denominator vanishes at 0; since $N(0) = 0$ the denominator there equals $-aD(0)$, which is nonzero for $a > 0$ and zero for $a = 0$. Hence the McMillan degree, which is the dimension of a minimal realization, is $1+|\mathrm{supp}(M)\cap(0,1)|$ for $a' \neq 0$ and $|\mathrm{supp}(M) \cap (0,1)|$ for $a' = 0$ (an atom at rate 0, a cohort with $\eta = 1$, contributes the constant $M(\{0\})$ to G and no pole, so it shifts a to $a' = a + \kappa M(\{0\})$ and the argument above applies with a') (numerically, the rank of the Hankel matrix of $(h_j)_{j\geq 1}$ is $K+1$ and K respectively for $K = 1, 2, 3$). For (ii), direct algebra gives

$$G_\delta(z) - G_0(z) = -\frac{qz(z-1)\delta^2}{(z-1+\bar\eta)\big((z-1+\bar\eta)^2 - \delta^2\big)}.$$

The denominator is the product of the three linear factors $z - r$ with $r \in \{1-\bar\eta,\, 1-\bar\eta\pm\delta\}$, all in $[\epsilon, 1-\epsilon]$ by assumption; on $|z| = 1$ each satisfies $|z - r| \geq 1 - r \geq \epsilon$, so $|G_\delta - G_0| \leq 2q\delta^2/\epsilon^3$ there. Uniform stability gives $M_H = \sup|H_\theta| < \infty$, and the identity $H_\delta - H_0 = z^{-1} H_\delta H_0 (G_\delta - G_0)$, whose factor z^{-1} has modulus one on the circle, gives an ℓ_2 gain at most $C_{\mathcal F}\delta^2$. The mean difference of a power-constrained probe has squared norm at most $C_{\mathcal F}^2\delta^4 PT$; the equal-noise KL is at most $C_{\mathcal F}^2\delta^4 PT/(2\sigma_\nu^2)$, and Pinsker's inequality with total variation at least $1 - 2\alpha$ gives (9).

Theorem 5.1. Let (s^A, ω^A) and (s^B, ω^B) be the K-point Gauss and Gauss–Radau rules for the uniform weight on $[-1, 1]$, normalized to unit mass. Both are positive and exact for polynomials of degree $2K - 2$, and their moments of degree $2K - 1$ differ by $g_K \neq 0$. To see that $g_K \neq 0$, let the Radau nodes be $-1, s_2, \ldots, s_K$ and put $P(s) = (s+1)\prod_{j\geq 2}(s - s_j)^2$. This is monic of degree $2K - 1$ and vanishes at every Radau node, so the Radau rule gives it 0. Gauss is exact to degree $2K - 1$, so it gives $\int P\, d\mu > 0$, since $P \geq 0$ on $[-1, 1]$ and vanishes only at the nodes. The two rules agree on every lower-degree term, so they differ on s^{2K-1}. Put $r = r_c + \Delta\eta\, s$ and weights $W\omega$, and let $\mu_j^X = W\sum_i \omega_i^X (s_i^X)^j$ for $X \in \{A, B\}$. Then $|\mu_j^X| \leq W$, and $\mu_j^A = \mu_j^B$ for $j \leq 2K - 2$.

Kernel gap. By the binomial theorem, $k_m^X = \sum_j \binom{m}{j} r_c^{m-j} (\Delta\eta)^j \mu_j^X$. So, with $n = 2K - 1$, $\Delta k_m = k_m^A - k_m^B = \sum_{j\geq n}\binom{m}{j} r_c^{m-j}(\Delta\eta)^j(\mu_j^A - \mu_j^B)$. Since $\binom{m}{j} \leq \binom{m}{n}\binom{m-n}{j-n}$ for $j \geq n$, this gives $|\Delta k_m| \leq 2W\binom{m}{n}(\Delta\eta)^n(r_c + \Delta\eta)^{m-n}$. Summing with $\sum_m \binom{m}{n} x^{m-n} = (1-x)^{-(n+1)}$ gives $\|\Delta k\|_1 \leq 2W(\Delta\eta)^{2K-1}/(1 - r_c - \Delta\eta)^{2K}$ over all lags.

Response gap. Let $\varsigma = a + \kappa W/(1 - r_c - \Delta\eta) < 1$, the theorem's hypothesis. Each kernel has $\|k^X\|_1 \leq W/(1 - r_c - \Delta\eta)$. From $h_0 = \kappa$ and $h_{t+1} = a h_t + \kappa \sum_j k_j h_{t-j}$, every partial sum of $|h^B|$ is at most $\kappa/(1-\varsigma)$. The difference $\Delta h = h^A - h^B$ has $\Delta h_0 = 0$ and $\Delta h_{t+1} = a\Delta h_t + \kappa\sum_j (k_j^A \Delta h_{t-j} + \Delta k_j h_{t-j}^B)$, so every partial sum of $|\Delta h|$ is at most $\kappa\|\Delta k\|_1\|h^B\|_1/(1-\varsigma)$. Hence, for every horizon,

$$\|h^A - h^B\|_2 \leq \|h^A - h^B\|_1 \leq C_U(\Delta\eta)^{2K-1}, \qquad C_U = \frac{2\kappa^2 W}{(1 - r_c - \Delta\eta)^{2K}(1-\varsigma)^2}.$$

In the other direction, $\Delta k_j = 0$ for $j < 2K - 1$, so by induction $\Delta h_t = 0$ for $t \leq 2K - 1$, and $\Delta h_{2K} = \kappa\, \Delta k_{2K-1} h_0^B = \kappa^2 W g_K (\Delta\eta)^{2K-1}$. When the taps $h_0, \ldots, h_{T-1}$ are observed with $T \geq 2K + 1$, this gives $\|h^A - h^B\|_2 \geq \kappa^2 W |g_K| (\Delta\eta)^{2K-1}$. With $T \leq 2K$ the observed responses coincide. The impossibility bound holds for every T, since it uses only the upper order.

Testing. N reset experiments have divergence $N\|h^A - h^B\|_2^2/(2\sigma^2)$. By Pinsker's inequality, a test whose two error probabilities are at most α needs divergence at least $2(1 - 2\alpha)^2$. So $N \geq 4(1-2\alpha)^2\sigma^2(\Delta\eta)^{-(4K-2)}/C_U^2$, and the constant does not depend on T. For the W_1 version, the

two merged measures are the images of the fixed normalized rules under $s \mapsto r_c + \Delta\eta\, s$ scaled to mass W, so their distance is $W\Delta\eta$ times the distance $2c_K > 0$ between the rules, and the same two-point argument applies with the metric loss.

Theorem 5.4. For stationary Gaussian processes with spectral densities $f_j \propto |H_j(e^{i\omega})|^2$, the divergence rate is

$$D = \frac{1}{4\pi}\int_{-\pi}^{\pi} (\varrho - 1 - \log\varrho)\, d\omega, \qquad \varrho = f_1/f_0 = |H_1/H_0|^2.$$

By the resolvent identity, $H_1/H_0 - 1 = z^{-1}H_1\Delta G$ with $\Delta G = G_1 - G_0$, and uniform stability bounds $|H_1|$ on the circle, so $u \equiv \varrho - 1 = 2\,\mathrm{Re}(z^{-1}H_1\Delta G) + O(|\Delta G|^2)$. Write $G(z) = \sum_k w_k \phi(z, \eta_k)$, with w_k the cohort weights (any factor of η_k is absorbed in ϕ), where ϕ is analytic in η on a neighbourhood of $[\eta_{\min}, \eta_{\max}]$ uniformly for $|z| = 1$ (its pole $z = 1 - \eta$ stays inside the disk). Expanding ϕ around $\bar\eta$, $\Delta G = \sum_{j\ge 0} \frac{\partial_\eta^j \phi(z,\bar\eta)}{j!}\, \Delta\tilde M_j$, where $\tilde M_j = \sum_k w_k(\eta_k - \bar\eta)^j$. For the pair, $\Delta\tilde M_0 = \Delta\tilde M_1 = 0$ and $\Delta\tilde M_2 = q\delta^2$; for the moment-matched economies, $\Delta\tilde M_j = 0$ for $j \le 2$ and $\Delta\tilde M_3 = 4\tau p_- p_+ (\Delta\eta)^3$ by (20); for the K-cohort Gauss and Gauss–Radau economies with rates $\bar\eta + \Delta\eta\, s$ and weights $W\omega$, both rules integrate polynomials of degree $2K-2$ exactly, so $\Delta\tilde M_j = W(\Delta\eta)^j \sum_k (\omega_k^A (s_k^A)^j - \omega_k^B (s_k^B)^j)$ vanishes for $j \le 2K-2$ and equals $\pm W g_K (\Delta\eta)^{2K-1} \ne 0$ at $j = 2K-1$. Higher terms are smaller by a factor of the spacing: $|\Delta\tilde M_j| \le 2W(\Delta\eta)^j$, and analyticity of ϕ in η on a disk of radius $\varrho > 0$ around $\bar\eta$, uniformly on the circle, gives $|\partial_\eta^j \phi|/j! \le C\varrho^{-j}$, so for $\Delta\eta < \varrho/2$ the tail beyond the first nonzero term is $O((\Delta\eta)^{j_0+1})$. Hence $\sup_\omega |u| = O(\delta^2)$, $O(\tau(\Delta\eta)^3)$, respectively $O((\Delta\eta)^{2K-1})$, and $u = c(\omega)\epsilon + o(\epsilon)$ with $\epsilon = \delta^2$, $\tau(\Delta\eta)^3$ or $(\Delta\eta)^{2K-1}$ and $c(\omega)$ proportional to $2\,\mathrm{Re}(z^{-1}H_0\,\partial_\eta^{j_0}\phi(\cdot,\bar\eta))/j_0!$, $j_0 = 2$, 3 or $2K-1$. The function c does not vanish identically: $f = z^{-1}H_0\partial_\eta^j\phi$ is analytic outside the disk including at infinity, where it vanishes, and is not identically zero; an analytic function on the exterior whose real part vanishes on the circle is an imaginary constant, which here would have to be zero. For $|u| \le \frac12$, $u^2/4 \le u - \log(1+u) \le u^2$, so $D \asymp \int c(\omega)^2 d\omega\, \epsilon^2$, which gives (i)–(iii). The sample requirements need a finite-T bound that is uniform in the separation, which we obtain directly. The first T observations are $N(0, \Sigma_j)$ with Σ_j the $T \times T$ Toeplitz matrices of the spectral densities $f_j = \sigma^2|H_j(e^{i\omega})|^2$. On the compact, uniformly stable family the transfer functions have no zeros or poles on the circle (Theorem 5.3), so $0 < \underline f \le f_j \le \bar f < \infty$ with constants that do not depend on the separation. A Toeplitz matrix has eigenvalues within the range of its symbol, so $\lambda_{\min}(\Sigma_0) \ge \underline f$ and $\|\Sigma_1 - \Sigma_0\| \le \sup_\omega |f_1 - f_0| = \sup_\omega f_0|u| \le \bar f \sup_\omega |u|$. With $A = \Sigma_0^{-1/2}\Sigma_1\Sigma_0^{-1/2}$, whose eigenvalues lie in $[\underline f/\bar f, \bar f/\underline f]$, the inequality $\lambda - 1 - \log\lambda \le (\lambda-1)^2/\lambda$ gives

$$D_{\mathrm{KL}}(P_1^T \| P_0^T) = \tfrac12 \sum_k (\lambda_k - 1 - \log\lambda_k) \le \frac{\bar f}{2\underline f}\, \|A - I\|_F^2 \le \frac{\bar f}{2\underline f}\, T \Big(\frac{\bar f}{\underline f}\Big)^2 \sup_\omega |u|^2 \le C\, T\epsilon^2,$$

with C independent of T and of the separation, because $\sup_\omega |u| = O(\epsilon)$ uniformly. The Bretagnolle–Huber inequality (Tsybakov, 2009, Lemma 2.6) then gives the stated lower bounds on T for every T, not only asymptotically. For a fixed pair, $T^{-1}D_{\mathrm{KL}}(P_1^T\|P_0^T) \to D$ and Stein's lemma (one error fixed) and the Chernoff bound (both errors vanishing) give the error exponents of the likelihood-ratio test, of the same order in the separation; that statement is pointwise in the pair.

Theorem 5.3. In the delay variable $\lambda = z^{-1}$, $H = \kappa \prod_i (1 - r_i\lambda) / \big[(1 - a\lambda)\prod_i (1 - r_i\lambda) - \kappa\lambda \sum_i w_i \prod_{j\ne i}(1 - r_j\lambda)\big]$, which is (4) divided through by z^{K+1}; its nonstructural zeros are the

memory rates, which do not cancel under the nondegenerate assumptions of Theorem 5.3. In the z-plane there is also the structural timing zero at zero: the numerator is proportional to $z\prod_i(z-r_i)$. Stability puts the poles inside the disk. A rational spectral density $\sigma^2|H(e^{i\omega})|^2$ has a unique minimum-phase factor up to a unimodular constant, which is H; Theorem 3.3 then gives the merged measure.

Proposition 6.2. With tagged flow, $u_{i,t} = r_i u_{i,t-1} + w_i x_t$ holds exactly and involves only cohort i, so $\tilde{u}_{i,t} = r_i\tilde{u}_{i,t-1} + w_i x_t + e_{i,t}$ with the MA(1) error $e_{i,t}$. Prices are independent of the measurement noise, so $\mathbb{E}[Z_t e_{i,t}] = 0$ and, by the ergodic theorem for the stationary linear price process, the IV estimator is consistent when $\Sigma_i = \mathbb{E}[Z_t(\tilde{u}_{i,t-1}, x_t)]$ is nonsingular. Since $u_{i,t-1} = w_i\sum_{s\geq 0} r_i^s x_{t-1-s}$, $\mathbb{E}[x_{t-1}u_{i,t-1}] = w_i A_0$ and $\mathbb{E}[x_t u_{i,t-1}] = w_i A_1$, which gives Σ_i. The price is a nonnegative mixture of geometric responses to its innovation (Lemma S1.1), so $\gamma_x(s) = \sum_\ell c_\ell\lambda_\ell^s$ with $c_\ell = \sigma_\xi^2\kappa^{-2}\omega_\ell\sum_m \omega_m/(1-\lambda_\ell\lambda_m) \geq 0$; substituting, $\det\Sigma_i = w_i(\gamma_x(0)A_0 - \gamma_x(1)A_1) = w_i\sum_{\ell,m} c_\ell c_m(1-\lambda_\ell\lambda_m)/(1-r_i\lambda_m)$, each term positive. By summation by parts, $\sum_{t=1}^T Z_t e_{i,t} = \sum_{t=1}^T(Z_t - r_i Z_{t+1})\varepsilon_{i,t} + O_P(1)$. Since $(\varepsilon_{i,t})$ is i.i.d. with mean zero and variance σ_f^2 and independent of the whole stationary price process, the summands $m_t = (Z_t - r_i Z_{t+1})\varepsilon_{i,t}$ form a stationary ergodic martingale difference sequence with respect to $\mathcal{F}_t = \sigma((x_s)_{s\in\mathbb{Z}}, \varepsilon_{i,s} : s \leq t)$, with $\mathbb{E}[m_t m_t^\top] = \sigma_f^2[(1+r_i^2)\Gamma_0 - r_i(\Gamma_1 + \Gamma_1^\top)] = \Omega_i$, so the Billingsley–Ibragimov martingale central limit theorem gives (11). On a uniformly stable family the modes and weights lie in a compact set with $\lambda_\ell \leq \bar{\lambda} < 1$, so with $w_i \geq w_{\min} > 0$ we get $\det\Sigma_i \geq w_{\min}(1-\bar{\lambda}^2)\gamma_x(0)^2 > 0$ and $\|\Omega_i\|$ is bounded, uniformly in the spacing.

Theorem 5.2.

Proof. The kernel values $k_m = \sum_i w_i r_i^m$ are the power moments of the merged measure. The closed-loop recursion $h_t = ah_{t-1} + \kappa\sum_m k_m h_{t-1-m}$ (plus κ at $t=0$) can be solved for $k_0,\dots,k_{2K-1}$ from $h_1,\dots,h_{2K}$. Because $h_0 = \kappa$ is a known primitive, each k_m is a polynomial in $h_1,\dots,h_{m+1}$ with coefficients depending on (a,κ) (the only division is by $\kappa h_0 = \kappa^2$), so on the bounded set of responses within sup-distance 1 of the class this map is L_h-Lipschitz, with L_h depending on the class but not on $\Delta\eta$. Let E be the event $\max_{1\leq t\leq 2K}|\bar{h}_t - h_t| \leq \sigma N^{-1/2}\sqrt{2\log(4K/\beta)}$, where $\bar{h}_t$ averages the N responses; a Gaussian tail bound and a union bound over the $2K$ estimated taps give $P(E) \geq 1-\beta$. On E, and provided this radius is at most 1, the estimated kernel values satisfy $\max_m|\hat{k}_m - k_m| \leq L_h\sigma N^{-1/2}\sqrt{2\log(4K/\beta)}$. Centring at r_c is a fixed triangular linear map, $\mu_m = \sum_{j\leq m}\binom{m}{j}(-r_c)^{m-j}k_j$, and $\mu_m = (\Delta\eta)^m t_m$ with $t_m = \sum_i w_i s_i^m$. The map $F : (s,\omega) \mapsto (t_0,\dots,t_{2K-1})$ does not depend on $\Delta\eta$; its Jacobian is a confluent Vandermonde matrix with determinant $\prod_i\omega_i\prod_{i<j}(s_i-s_j)^4$, bounded away from zero on the compact class. On ordered, γ-separated nodes F is injective, because a measure with K distinct atoms is determined by its first $2K$ moments. A C^1 map that is injective on a compact set Q and has a nonsingular Jacobian there is injective on an open neighbourhood U of Q: otherwise there are $x_n \neq y_n$ with $F(x_n) = F(y_n)$ and distance to Q tending to zero, whose limits $x, y \in Q$ satisfy $F(x) = F(y)$, so $x = y$, contradicting local injectivity at x. Applying the quantitative inverse function theorem uniformly on Q, there are $\rho_0 > 0$ and $L < \infty$, uniform over the class, such that $F(U)$ contains the ball of radius ρ_0 around every point of $F(Q)$ and the single inverse $F^{-1} : F(U) \to U$ is L-Lipschitz on each such ball. An error e in μ_m is an error $e(\Delta\eta)^{-m}$ in t_m, largest at $m = 2K-1$. Since $\Delta\eta < 1$ and the centring map is fixed, on E we get $\|\hat{t} - t\|_\infty \leq C_1\sqrt{\log(4K/\beta)}\,\epsilon_N$ with C_1 equal to $\sqrt{2}L_h$ times the norm of the centring map; and because $\sigma N^{-1/2}\sqrt{2\log(4K/\beta)} \leq \sqrt{2}\,\epsilon_N\sqrt{\log(4K/\beta)}$, the radius condition holds whenever $c_0 \leq 1/\sqrt{2}$. When this is below ρ_0, $\hat{t}$ lies in the ball where F^{-1}

is L-Lipschitz. For an input in such a ball, $x = F^{-1}(\hat{t})$ has K distinct nodes and nonzero weights, so the $K \times K$ Hankel matrix of $\hat{t}$ is nonsingular and Prony's method returns exactly x: the Prony polynomial is determined uniquely by the Hankel system, and since the K real nodes of x reproduce $\hat{t}$ it equals $\prod_i (z - \hat{s}_i)$, so its roots are real and distinct and no complex pair can appear; Prony's method computes the algebraic inverse of the moment map up to the order of the atoms, so, listing its output in increasing order of rate (which the error bound below keeps unambiguous once $C\sqrt{\log(4K/\beta)}\,\epsilon_N < \gamma/2$), its output is $F^{-1}(\hat{t})$ and the stated bound follows with $C = LC_1$ and $c_0 = \min\{\min(\rho_0, \gamma/(2L))/C_1,\, 1/\sqrt{2}\}$, which also keeps $C\sqrt{\log(4K/\beta)}\,\epsilon_N < \gamma/2$. No separate root-perturbation bound is needed: on the ball where F^{-1} is L-Lipschitz, Prony's roots and weights are the coordinates of $F^{-1}(\hat{t})$, so the conditioning of the root step is part of L (for an explicit step-by-step analysis of Prony's method with clustered exponents see Katz et al., 2023). The matching statement requires the class to contain the Gauss and Gauss–Radau pair of Theorem 5.1 (ordered nodes with separation at least the smaller of their two minimum gaps, and weights within $[\omega_{\min}, \omega_{\max}]$ after scaling to total weight W); for any such class the orders match, with constants that depend on the class. □

Theorem 6.3.

Proof. $N_p \asymp \sigma^2(\Delta\eta)^{-(4K-2)}$ by Theorems 5.1 and 5.2. *Upper bounds.* Average the N price responses to estimate h with error of order $\sigma N^{-1/2}$ per tap. Given h, class i's mean flow is a two-parameter family $F(w_i, r_i; h)$ whose Jacobian columns $\sum_s r_i^{t-s} h_s$ and $w_i \sum_s (t-s) r_i^{t-s-1} h_s$ are linearly independent, with a Gram determinant bounded below uniformly on the class because $w_i \geq \omega_{\min}$, the rates lie in a compact subset of $(0,1)$ and $h_0 = \kappa > 0$; the bound does not involve the other classes or $\Delta\eta$. Least squares on class i's flows with the plug-in h therefore estimates (w_i, r_i) with error $O_P(N^{-1/2})$, uniformly on the class, which gives the upper bounds in (i) and in (ii) for accuracy $\epsilon\Delta\eta$. For (i) there is also a finite-sample version: in the first period the tagged flow of class i is $w_i\kappa$, so the mean of the N first-period records gives $P(|\hat{w}_i - w_i| \geq \epsilon) \leq \sigma_f^2/(N\kappa^2\epsilon^2)$ by Chebyshev's inequality, for every configuration (this step is machine-checked). *Lower bounds.* At a configuration in the interior of the class, perturb one class's weight by ϵ, or its rate by $\epsilon\Delta\eta$; for small ϵ both perturbed configurations remain in the class because the weight interval is nondegenerate and the rate set has nonempty interior. The mean differences of both the price and the flow records are of order ϵ, respectively $\epsilon\Delta\eta$, so the Kullback–Leibler divergence of N experiments is at most $CN\epsilon^2$, respectively $CN\epsilon^2(\Delta\eta)^2$, and Le Cam's two-point lemma gives the lower bounds. □

Unknown primitives.

Corollary S1.2 (Unknown primitives cost nothing in the rate)**.** *In the setting of Theorem 5.2, suppose in addition that every memory rate is at least $\varepsilon > 0$ (that is, $r_c - \Delta\eta \geq \varepsilon$), and that the fundamental value p^*, the impact $\kappa \in [\underline{\kappa}, \overline{\kappa}] \subset (0, \infty)$ and the persistence $a \in [0, \bar{a}] \subset [0, 1)$ are unknown. Put $\tilde{k}_0 = k_0 + a/\kappa$ and $\tilde{k}_m = k_m$ for $m \geq 1$.*

(i) The response depends on (a, κ, M) only through κ and $\tilde{k}$: $h_0 = \kappa$ and $h_{t+1} = \kappa \sum_{m=0}^{t} \tilde{k}_m h_{t-m}$. The persistence a acts exactly like an extra group with memory rate zero and merged weight a/κ.

(ii) The taps $h_0, \ldots, h_{2K+1}$ identify (κ, a, M): $\kappa = h_0$; the recursion gives $\tilde{k}_0, \ldots, \tilde{k}_{2K}$; the values $\tilde{k}_1, \ldots, \tilde{k}_{2K}$ are the moments of orders $0, \ldots, 2K-1$ of $\mu' = \sum_i w_i r_i \, \delta_{r_i}$; and $a = \kappa(\tilde{k}_0 - \sum_i w_i)$.

(iii) Subtract the pre-probe price to remove p^* (doubling the noise variance), estimate $\hat{\kappa} = \hat{h}_0$, deconvolve with $\hat{\kappa}$ alone, apply Prony's method to the estimated moments of μ', and set

$\hat{w}_i = \widehat{w_i r_i}/\hat{r}_i$ and $\hat{a} = \hat{\kappa}(\hat{\tilde{k}}_0 - \sum_i \hat{w}_i)$. There are constants $c_0, C > 0$ depending on $(K, \mathcal{S}, \gamma, \omega_{\min}, \omega_{\max}, \varepsilon, \underline{\kappa}, \overline{\kappa}, \bar{a}, r_c)$ but not on $\Delta\eta$ such that, whenever $\epsilon_N \le c_0$,

$$\max_i |\hat{w}_i - w_i| + |\hat{a} - a| + |\hat{\kappa} - \kappa| \;\le\; C\sqrt{\log(8K/\beta)}\,\epsilon_N$$

with probability at least $1-\beta$. Hence $N = C'\sigma^2(\Delta\eta)^{-(4K-2)}$ experiments suffice, the same order as with known primitives.

In words. Not knowing the market's own persistence or its price impact costs a constant factor, not a worse rate. Persistence enters the price exactly like one more group of traders with no memory at all, and that group sits far from the clustered groups the supervisor is trying to separate. Measured in the merged weights times the rates, the clustered groups' problem is the same Prony problem as before, with weights bounded below because rates are bounded away from zero; one more tap pins down the extra group.

Proof of Corollary S1.2. (i) Substitute $k_0 = \tilde{k}_0 - a/\kappa$ in $h_{t+1} = ah_t + \kappa\sum_{m\le t} k_m h_{t-m}$: the term ah_t cancels against $\kappa(a/\kappa)h_t$. (ii) Solving the recursion of (i) for $\tilde{k}_t$ gives $\tilde{k}_t = (h_{t+1} - \kappa\sum_{m<t}\tilde{k}_m h_{t-m})/\kappa^2$, a polynomial in $h_1, \dots, h_{t+1}$ divided by a power of $h_0 = \kappa$. For $m \ge 1$, $\tilde{k}_m = \sum_i w_i r_i^m = \sum_i (w_i r_i) r_i^{m-1}$, so $\tilde{k}_1, \dots, \tilde{k}_{2K}$ are the moments of orders $0, \dots, 2K-1$ of μ', which has K distinct atoms with positive weights; these moments determine μ' (Theorem 3.3). Since $r_i \ge \varepsilon > 0$, $w_i = (w_i r_i)/r_i$, and then $a = \kappa(\tilde{k}_0 - \sum_i w_i)$. (iii) On the event that every estimated tap is within $\sigma N^{-1/2}\sqrt{2\log(8K/\beta)}$ of its mean (probability at least $1-\beta/2$ by the Gaussian tail and a union bound over $2K+2$ taps, with the doubled variance absorbed into the constant), $\hat{\kappa} \ge \underline{\kappa}/2$ once $\epsilon_N \le c_0$, so the map from taps to $\hat{\tilde{k}}$ is Lipschitz with a constant that depends on $(K, \underline{\kappa}, \overline{\kappa}, \bar{a}, W)$ and not on $\Delta\eta$. The weights of μ' lie in $[\varepsilon\omega_{\min}, \omega_{\max}]$ and its rates are those of the class of Theorem 5.2; the proof of that theorem uses only a $\Delta\eta$-free Lipschitz deconvolution, the rescaling by $(\Delta\eta)^{-m}$ and weights bounded below, so it applies verbatim to μ' and gives $\max_i |\widehat{w_i r_i} - w_i r_i| \le C_1\sqrt{\log(8K/\beta)}\,\epsilon_N$ and rate errors smaller by a factor $\Delta\eta$, with probability at least $1-\beta/2$. The maps $(v, r) \mapsto v/r$ on $r \ge \varepsilon/2$ and $(\kappa, \tilde{k}_0, w) \mapsto \kappa(\tilde{k}_0 - \sum_i w_i)$ are Lipschitz on the bounded set reached, and $|\hat{\kappa} - \kappa|$ and $|\hat{\tilde{k}}_0 - \tilde{k}_0|$ are of order $\sigma N^{-1/2} \le \epsilon_N$. Combining gives the bound. □

Proposition 3.2. The proof is given after the statement in the main text.

S2 Equilibrium: further results and proofs

Corollary S2.1 (Disclosure in equilibrium)**.** *Take a positive-impact equilibrium of Theorem 4.1 with positive weights and $D > \max_i r_i$.*

(i) The cohorts' flows obey $u_{i,t+1} = r_i u_{i,t} + w_i x_t$ with their true *weights. The price is the stable closed loop of Theorem 4.2, driven by ξ, with positive effective weights. So Proposition 6.2 applies: tagged flow identifies each cohort's (r_i, w_i), with a limiting covariance that has no singularity as rates collide.*

(ii) On classes in which both the true and the effective weights lie in compact positive intervals, Theorem 6.3 holds in equilibrium. Tagging cuts the experiments needed by a factor of order $(\Delta\eta)^{-(4K-2)}$ for weights and $(\Delta\eta)^{-(4K-4)}$ for rates.

In words. What makes disclosure valuable survives the dealers' optimization. Prices still hide the groups at the same polynomial cost, and each group's own flow still reveals that group's memory and weight directly.

Theorem S2.2 (Fragility: where liquidity disappears)**.** *Let there be one feedback cohort with weight $w > 0$ and memory rate $r \in (0,1)$, with $\bar{a} \in [0,1)$ and $\sigma_F, \sigma_\xi > 0$. Call an equilibrium of Theorem 4.1 with $\kappa > 0$ and $D > 0$ a positive-impact equilibrium. For $D \in (m,1)$ with $m = \max(\bar{a}, r)$, let*

$$\kappa_w(D) = \frac{(1-D)(D-r)}{w}, \qquad \Psi_w(D) = \frac{\kappa_w(D)\,(D-\bar{a})\,D}{\sigma_F^2 D^2 + \kappa_w(D)^2 \sigma_\xi^2}, \qquad A^*(w) = \max_{D \in (m,1)} \Psi_w(D).$$

(i) Threshold.

- *The positive-impact equilibria are exactly the pairs $(\kappa_w(D), D)$ with $D \in (m,1)$ and $\Psi_w(D) = A$, and every one of them is stable.*
- *A positive-impact equilibrium exists if and only if $A \le A^*(w)$.*
- *For $A < A^*(w)$ there are at least two; for $A > A^*(w)$ there are none.*

(ii) Monotonicity. *$A^*(w)$ is strictly decreasing in w. As $w \downarrow 0$ it tends to $A_0^* = (1-\bar{a})/(2\sigma_F\sigma_\xi)$, the threshold of Theorem 4.1(ii).*

(iii) Bounds and rate. *For every $w > 0$,*

$$A^*(w) < \min\Big\{A_0^*,\ \frac{\Phi^*}{\sigma_F^2\, w}\Big\},$$

where

$$\Phi^* = \max_{D \in [m,1]} \frac{(1-D)(D-r)(D-\bar{a})}{D} \le \frac{(1-\bar{a})(1-r)^2}{4},$$

and $w\,A^*(w) \to \Phi^*/\sigma_F^2$ as $w \to \infty$. In terms of the feedback intensity $G(1) = w/(1-r)$, strong feedback gives $A^* \approx \Phi^*/(\sigma_F^2(1-r)G(1))$, and always $A^* < (1-\bar{a})(1-r)/(4\sigma_F^2 G(1))$.

(iv) Several cohorts. *Let all weights be positive.*

- *The positive-impact equilibria with $D > \max_i r_i$ are exactly the pairs $((1-D)/G(D), D)$ with $D \in (\max(\bar{a}, \max_i r_i), 1)$ at which the analogue of Ψ_w, with $\kappa = (1-D)/G(D)$, equals A.*
- *Every such equilibrium has*

$$A < \min\Big\{A_0^*,\ \frac{(1-\sqrt{\bar{a}})^2}{\sigma_F^2\, G(1)}\Big\},$$

  whatever the number of cohorts and their rates.
- *Every such equilibrium is stable.*

In words. With positive feedback, dealers can make a market only if their dislike of risk is below a threshold, and the threshold falls as feedback grows: for strong feedback it is inversely proportional to the feedback intensity. Past the threshold, no price impact is consistent with dealers' optimization. Below it, every equilibrium has stable prices. Liquidity does not erode through growing instability; it disappears when the two equilibria merge, at a point that stable

prices give no warning of. At the calibration of Supplement S2 ($\bar{a} = 0.6$, $r = 0.7$, $\sigma_F^2 = 1$, $\sigma_\xi^2 = 0.25$), A^* is 0.40 without feedback, 0.25 at $G(1) = \frac{1}{15}$ and 0.034 at $G(1) = \frac{2}{3}$, and $w\,A^*$ approaches $\Phi^* = 0.00688$. The necessary conditions behind parts (i) and (iii) are machine-checked (`Paper1EquilibriumReducedForm.lean`). With several cohorts, part (iv) gives a threshold that depends on the population only through its total feedback intensity $G(1)$. At the same calibration, numerically, the exact thresholds for two and three cohorts lie between 28% and 86% of that bound, and the equilibria on the threshold curve are stable. Every such equilibrium is stable (part (iv)), as the scan also shows.

Theorem S2.3 (A strategic insider in the inventory equilibrium)**.** *Add three things to the model of Theorem 4.1:*

- News. *The fundamental moves by $F_{t+1} = F_t + \theta_t + \varepsilon_{t+1}$, with $\theta_t \sim N(0, \sigma_\theta^2)$ and ε_t i.i.d. of variance σ_ε^2.*
- An insider. *At date t a short-lived insider privately observes θ_t, and nothing else, submits d_t, and values the position at F_{t+1}.*
- What cohorts and dealers see. *The cohorts trade on the observed deviation $s_t = p_t - F_t$. The dealers price from total flow.*

Let $\nu_t = \xi_t + d_t$ be the unexpected flow.

(i) Characterization. *The strategy $d_t = b\theta_t$ and the price $p_t = F_t + \lambda\nu_t + x_t$, with x_t the price law of Theorem 4.1, form an equilibrium exactly when the induced dynamics are stable and*

$$D = 1 - \kappa G(D), \quad \kappa(D - \bar{a}) = AVD, \quad \lambda = \frac{b\sigma_\theta^2}{\sigma_\nu^2}, \quad b = \frac{D}{2(\kappa + \lambda)},$$

$$\sigma_\nu^2 = \sigma_\xi^2 + b^2\sigma_\theta^2, \qquad V = \sigma_\varepsilon^2 + \frac{\sigma_\theta^2\sigma_\xi^2}{\sigma_\nu^2} + \Big(\frac{\kappa + \lambda}{D}\Big)^2\sigma_\nu^2.$$

One unit of unexpected flow moves the price by $(\kappa + \lambda)/D$, and the insider's expected profit per date is $b\sigma_\theta^2/2$.

(ii) Kyle limit. *Without inventory risk ($A = 0$), $\kappa = 0$, $D = 1$, $\lambda = \sigma_\theta/(2\sigma_\xi)$ and $b = \sigma_\xi/\sigma_\theta$. These are Kyle's one-period values at every date, and the feedback cohorts do not move prices.*

(iii) Invariance. *The coefficients depend on the cohorts only through G, as Theorem 3.1 requires.*

(iv) Observer. *The observed deviation follows*

$$s_t = a^* s_{t-1} + \kappa^* \sum_i \tilde{u}_{i,t} + c_0\nu_t - c_1\nu_{t-1}, \qquad c_0 = \frac{\kappa + \lambda}{D}, \quad c_1 = \frac{\bar{a}\lambda}{D},$$

with $(a^*, \kappa^*, \tilde{u})$ as in Theorem 4.2. This is the closed loop of Section 2 driven through a known moving-average filter $c_0 - c_1 z^{-1}$, which is invertible because $c_1 < c_0$.

- *The memory rates remain the transmission zeros of the feedback part.*
- *The lower-bound pair of Theorem 5.1 is realized by two such equilibria with identical primitives.*
- *So the identification boundary and the $(\Delta\eta)^{-(4K-2)}$ rate hold with an insider present.*

In words. A short-lived insider who knows the next piece of news trades against noise traders and against the feedback traders' predictable flow. Dealers net the predictable flow out. So the insider's trades move prices through two channels: an information part λ and an inventory part, both amplified by feedback through D. Positive feedback raises the total impact. The insider then trades less and earns less, and prices reveal less of her news. Negative feedback does the opposite.

- *An example from Supplement S2.* One positive-feedback cohort lowers the share of the insider's information impounded in prices from 0.372 to 0.366. One negative-feedback cohort raises it to 0.415.
- *What prices cannot identify.* It is still worth nothing to anyone trading on public information.
- *Machine check.* `Paper1EquilibriumInsider.lean` checks the observer's pathwise reduced form, the insider's best response, the Kyle limit and the posterior variance.
- *Scope.* The insider's information is short-lived, and she conditions only on her news. Two cases are not solved: an insider who also saw the public state, who would trade against the predictable mispricing; and a long-lived Kyle insider, who would also time her trades against the cohorts' memory.

A misjudged dealer and a targeted halt. The next two results use the simplest linear market inside Theorem 3.1. They are summarized in Section 4.

- Dates are $t = 0, \dots, T-1$ and the price change is $s_t = p_t - p_{t-1}$, with $p_{-1} = 0$.
- The dealer moves the price by $s_t = \lambda(y_t - \hat{f}_t)$, with $\lambda > 0$, where $\hat{f}_t = \sum_{m<t} \hat{k}_m s_{t-1-m}$ is the feedback flow it predicts from its kernel $\hat{k}$.
- The insider, with value v of mean zero and variance Σ_0 independent of the noise, commits to a schedule $x = bv$, $b \in \mathbb{R}^T$. Schedules may change sign, so pump-and-dump patterns are allowed.
- The kernel error is $\Delta k = k - \hat{k}$, with $\varepsilon = \sum_{m \le T-2} |\Delta k_m|$.

Let C be the lower-triangular matrix of ones and L the strictly lower Toeplitz matrix with $L_{t,t-1-m} = \Delta k_m$.

Theorem S2.4 (Rents from a misjudged aggregate response)**.** *(i) The insider's expected profit is $J(b) = \Sigma_0(\mathbf{1}^\top b - b^\top A b)$, with $A = \lambda C(I - \lambda L)^{-1}$. If $Q = \frac{1}{2}(A + A^\top) \succ 0$, the optimal schedule is $b^* = \frac{1}{2}Q^{-1}\mathbf{1}$ and the value is $V(\Delta k) = \frac{\Sigma_0}{4}\mathbf{1}^\top Q^{-1}\mathbf{1}$.*

(ii) If $\lambda T \varepsilon \le \frac{1}{4}(1 - \lambda\varepsilon)$, then $Q \succ 0$ and

$$|V(\Delta k) - V(0)| \le \frac{2\Sigma_0 T^2 \varepsilon}{1 - \lambda\varepsilon}.$$

(iii) To first order, $V(\Delta k) = V(0) - \Sigma_0 \sum_m \omega_m \Delta k_m + O(\varepsilon^2)$, with $\omega_m = \frac{(T-m)(T-m-1)}{2(T+1)^2} > 0$ and $V(0) = \frac{\Sigma_0 T}{2\lambda(T+1)}$. The rent is therefore of exact order ε. The insider gains when the dealer overstates the feedback ($\Delta k < 0$) and loses when it understates it.

(iv) If the dealer estimates the aggregate kernel from n observations with error $\varepsilon_n = O_p(n^{-1/2})$, the rent is $O_p(n^{-1/2})$. Separating the groups behind the same kernel still needs order $(\Delta\eta)^{-(4K-2)}$ experiments (Theorem 5.1).

In words. A dealer that misjudges how strongly feedback traders respond hands the insider a rent, or charges it one, in proportion to the error. That error concerns only the aggregate response, which public data reveal at the ordinary rate. So the information that protects price formation is cheap to learn, while the information about who the groups are stays expensive. The dealer's coefficient λ is held fixed: this is the insider's best response to a misspecified dealer, not a re-solved equilibrium.

Theorem S2.5 (Targeted intervention in equilibrium)**.** *Take the Gauss and Radau pairs of Theorem 3.1(iv), with total weight W, on rates $c+ds$ with $\rho = |c|+|d| < 1$. Place them in the market above, with the same dealer rule $(\lambda, \hat{k})$, the same insider schedule and the same draws.*

(i) The kernels agree for $m \le 2$, and $|k^G_m - k^R_m| \le 2W\binom{m}{3}|d|^3\rho^{m-3}$ for $m \ge 3$. Hence $\|k^G - k^R\|_1 \le 2W|d|^3/(1-\rho)^4$.

(ii) Before any intervention, the price paths differ by at most $\lambda^2 T\|k^G - k^R\|_1\|x+z\|$ divided by $(1-\lambda\|k^G-\hat{k}\|_1)(1-\lambda\|k^R-\hat{k}\|_1)$, which is $O(|d|^3)$. The insider's values differ by $O(|d|^3)$.

(iii) Halting the largest cohort leaves kernels $\frac{W}{2}(c-d\sigma)^m$ (the Gauss weights tie; we remove the cohort at $c+d\sigma$) and $\frac{W}{4}(c-d)^m$, which differ by $\frac{W}{4}$ at $m=0$ for every spacing. The post-halt responses therefore differ at order one, and so do the post-halt feedback budgets $\sum_m k_m$.

In words. Two markets can look the same in equilibrium to within the cube of the memory spacing, yet halting their largest group leaves feedback that differs at order one. Equilibrium does not remove the case for targeted information; it sharpens it. In the simulation below, the pre-halt price gap is about $3d^3$ at all three spacings tried, while the post-halt gap stays near 0.19.

Proof of Theorem 3.1. Part (i). By induction, $m_{i,t} = \sum_{m<t} r_i^m s_{t-1-m}$. Multiplying by w_i and summing over i gives $f_t = \sum_{m<t} k_m s_{t-1-m}$.

Part (ii). If $k_m = k'_m$ for $m < T$, then by (i) the two aggregate flows agree on $t \le T$ for every signal. Conversely, the impulse signal sequence $s_j = \mathbf{1}\{j=0\}$ gives $f_{m+1} = k_m$, so equal flows on $t \le T$ for every signal sequence force $k_m = k'_m$ for $m < T$. This direction quantifies over signal sequences, not over price paths under a fixed rule S: a rule such as $S \equiv 0$ never produces the impulse.

On an unbounded horizon, equality of all $k_m = \int r^m\, dM(r)$ is equality of all moments of two finite signed measures on $[0,1]$. Polynomials are dense in $C[0,1]$, so the measures coincide. In terms of η this is the condition (8) of Theorem 3.3.

Part (iii). Fix a strategy profile and a realization. We argue by strong induction on $t \le T$. If prices and flows agree before t:

- the signals agree before t, because S_j depends on prices up to j;
- the aggregate flows at t agree, by part (i) and the equal kernels, since f_t uses signals before t only;
- the insider's orders at t agree, by causality;
- hence the total flows y_t agree, and then the prices p_t agree, by causality of the dealer's rule.

The insider's profit $\sum_{t<T}(v-p_t)x_t$ and the dealer's inventory $-\sum_{t<T} y_t$ are functions of these paths.

Strategy sets are the same in the two games, because the rules are defined on the same histories. Every payoff is an expectation of the same function of (v,z), so best responses coincide and so do the equilibria.

Part (iv). The Gauss rule has weights $\frac{1}{2}, \frac{1}{2}$ and nodes $\pm\sigma$ with $\sigma^2 = \frac{1}{3}$. The Radau rule has weights $\frac{1}{4}, \frac{3}{4}$ and nodes $-1, \frac{1}{3}$. Both integrate $1, s, s^2$ exactly, so on rates $c + ds$ the kernels agree for $m \le 2$. The largest weights are $\frac{1}{2}$ and $\frac{3}{4}$. □

Machine check. `Paper1EquilibriumInvariance.lean` proves the following:

- part (i): `p1_eqAggFlow_kernel`;
- both directions of part (ii) on a finite horizon: `p1_eqAggFlow_of_kernel` and `p1_kernel_of_eqAggFlow`;
- the pathwise statement of part (iii) for arbitrary causal insider, dealer and signal rules: `p1_eq_market_invariance` and `p1_eq_insider_profit_invariance`;
- part (iv): `p1_eq_group_level_differs`.

The unbounded-horizon moment statement and the passage from pathwise equality to equal equilibria are on paper.

Cohort-specific demand shocks. Suppose cohort i also trades an idiosyncratic shock $\epsilon_{i,t}$ through its memory, so that f_t gains the term $\sum_i w_i \sum_{m<t} r_i^m \epsilon_{i,t-1-m}$. The same induction shows that the equilibrium depends on the cohorts only through the kernel and the law of this aggregate shock process. For independent Gaussian shocks with variances σ_i^2, that law is determined by the spectral density $\sum_i w_i^2 \sigma_i^2 |1 - r_i e^{i\omega}|^{-2}$. This is a second merged object, and the invariance class shrinks accordingly. The paper's cohorts have no idiosyncratic shocks.

The textbook benchmark. Take a risk-neutral, competitive dealer that knows the kernel. It can compute f_t from public prices and price only $y_t - f_t$, so predictable feedback flow then has no price impact, and the closed loop of Section 2 is absent in that equilibrium. The theorem holds in that benchmark as well. Feedback reaches prices in equilibrium when the dealer bears inventory risk, or when it does not know the kernel and must learn it. These are the cases studied in the equilibrium extension.

Numerical illustration. `equilibrium/e1_demo.py` runs one market. The insider submits $x_t = 0.3(v - p_{t-1})$. The dealer sets $p_t = p_{t-1} + 0.3\, y_t - 0.05\,(I_t - I_{t-1})$. The signal is $s_t = p_t - p_{t-1}$. Two cohort configurations are compared, the Gauss and Radau pairs of part (iv) on rates $0.6 + 0.2s$, with total weight 0.5, under the same (v, z) draws. Over 200 draws:

- price paths coincide to machine precision on the first four dates ($t \le 3$), with largest gap 2×10^{-16};
- from date five, where k_3 first differs (by $\frac{2}{9}d^3$ times the total weight), they separate, with largest gap 1.8×10^{-4} at date five and 1.9×10^{-2} by date twenty;
- the largest group's share of feedback positions stays at medians of 0.59 and 0.79.

The result file is `equilibrium/e1_demo_results.json`.

Proof of Theorem S2.4. *Part (i).* Total flow is $y = x + z + f$. The dealer subtracts $\hat{f}$, so $s = \lambda(x + z + Ls)$. The matrix L is strictly lower triangular, so $I - \lambda L$ is invertible and $s = \lambda(I - \lambda L)^{-1}(x + z)$. Prices are $p = Cs$. With $x = bv$, realized profit is $\sum_t (v - p_t)x_t = v^2 \mathbf{1}^\top b - v\, b^\top C s$. Taking expectations with $E[v] = 0$ and v independent of z gives $J(b)$. When $Q \succ 0$, J is strictly concave, and its maximizer and value are as stated.

Part (ii). A Toeplitz operator is bounded by the ℓ_1 norm of its symbol, so $\|L\| \le \varepsilon$. For $\lambda\varepsilon < 1$ this gives $\|(I-\lambda L)^{-1}\| \le 1/(1-\lambda\varepsilon)$ and $(I-\lambda L)^{-1} - I = \lambda L(I-\lambda L)^{-1}$. Since $\|C\| \le T$, we get $\|Q - Q_0\| \le \|A - A_0\| \le \lambda^2 T\varepsilon/(1-\lambda\varepsilon)$. Moreover $Q_0 = \frac{\lambda}{2}(I + \mathbf{1}\mathbf{1}^\top) \succeq \frac{\lambda}{2} I$. Under the stated condition $\|Q - Q_0\| \le \lambda/4$, so $Q \succeq \frac{\lambda}{4} I$. Then

$$|V - V_0| = \tfrac{\Sigma_0}{4} |\mathbf{1}^\top Q^{-1}(Q_0 - Q)Q_0^{-1}\mathbf{1}| \le \tfrac{\Sigma_0}{4}\, T \cdot \tfrac{4}{\lambda} \cdot \tfrac{2}{\lambda} \cdot \tfrac{\lambda^2 T\varepsilon}{1-\lambda\varepsilon}.$$

Part (iii). $Q_0^{-1}\mathbf{1} = \frac{2}{\lambda(T+1)}\mathbf{1}$, which gives $V(0)$. Since $A = A_0 + \lambda^2 CD + O(\varepsilon^2)$, the derivative of $\mathbf{1}^\top Q^{-1}\mathbf{1}$ is $-u^\top(\lambda^2 CD)u$ with $u = Q_0^{-1}\mathbf{1}$. Using $\mathbf{1}^\top C = (T, T-1, \dots, 1)$ and $(L\mathbf{1})_t = \sum_{m<t} \Delta k_m$, we get $\mathbf{1}^\top CL\mathbf{1} = \sum_m \Delta k_m \frac{(T-m)(T-m-1)}{2}$. Multiplying by $\frac{\Sigma_0}{4} \cdot \frac{4}{\lambda^2(T+1)^2} \cdot \lambda^2$ gives ω_m.

Part (iv). This is part (ii) with $\varepsilon = \varepsilon_n$. The rate for ε_n is an assumption on the dealer's estimator, for example least squares of order flow on lagged price changes in a stationary market. It is not proved here.

Proof of Theorem S2.5. *Part (i).* Write μ_j^G, μ_j^R for the jth moments of the two rules, which are at most one in absolute value. Expanding $(c + ds)^m$ gives

$$k_m^G - k_m^R = W \sum_{j \ge 3} \binom{m}{j} c^{m-j} d^j (\mu_j^G - \mu_j^R).$$

The terms with $j \le 2$ vanish because both rules are exact for those degrees. Since $\binom{m}{j} \le \binom{m}{3}\binom{m-3}{j-3}$, the sum is at most $2W\binom{m}{3}|d|^3\rho^{m-3}$. Finally, $\sum_{m\ge 3}\binom{m}{3}\rho^{m-3} = (1-\rho)^{-4}$.

Part (ii). With residual matrices L_G, L_R built from $k^G - \hat{k}$ and $k^R - \hat{k}$,

$$\begin{aligned} p^G - p^R &= \lambda C\big[(I-\lambda L_G)^{-1} - (I-\lambda L_R)^{-1}\big](x+z) \\ &= \lambda^2 C(I-\lambda L_G)^{-1}(L_G - L_R)(I-\lambda L_R)^{-1}(x+z). \end{aligned}$$

Bound each factor as in the proof of Theorem S2.4. The value comparison follows from the same perturbation argument, with base point L_R instead of 0.

Part (iii). Removing one of the two Gauss cohorts leaves weight $W/2$. Removing the Radau cohort with weight $3/4$ leaves $W/4$. At $m = 0$ the kernel is the remaining weight.

Proof of Theorem 4.1. *Part (i).* Conjecture $x_t = -\kappa H_t + \sum_i \beta_i u_{i,t+1}$ and set $D = 1 - \sum_i \beta_i w_i$. The next-period expectation follows from three facts: $E_t\xi_{t+1} = 0$, $E_t H_{t+1} = \bar{a} H_t - \sum_i u_{i,t+1}$, and $E_t u_{i,t+2} = r_i u_{i,t+1} + w_i E_t x_{t+1}$. Together they give

$$D\, E_t x_{t+1} = -\kappa\bar{a} H_t + \sum_i (\kappa + \beta_i r_i) u_{i,t+1}.$$

The dealers' condition $x_t = E_t x_{t+1} - AVH_t$ must hold for all states. Matching the coefficient on $u_{i,t+1}$ gives $\beta_i = (\kappa + \beta_i r_i)/D$, that is, $\beta_i = \kappa/(D - r_i)$. The definition of D then becomes $D = 1 - \kappa G(D)$, and matching on H_t gives $\kappa(D - \bar{a}) = AVD$.

For the variance, x_{t+1} depends on ξ_{t+1} through H_{t+1} and through its own effect on $u_{i,t+2}$. Its innovation δ therefore satisfies $\delta = \kappa\xi_{t+1} + \sum_i \beta_i w_i \delta$, so $\delta = \kappa\xi_{t+1}/D$ and $V = \sigma_F^2 + \kappa^2\sigma_\xi^2/D^2$.

Conversely, take a solution of (10). Then the conjectured law satisfies the dealers' condition in every state (`p1_inventory_foc`), and V is consistent with it (`p1_inventory_innovation`). If the induced dynamics are stable, the process is stationary and the solution is an equilibrium.

Part (ii). With no cohorts, β is empty and $D = 1$. Then $x_t = -\kappa H_t$ and $H_t = \bar{a}H_{t-1} - y_t$ give $x_t = \bar{a}x_{t-1} + \kappa y_t$. The quadratic $A\sigma_\xi^2\kappa^2 - (1-\bar{a})\kappa + A\sigma_F^2 = 0$ has discriminant Δ (`p1_inventory_benchmark`).

Part (iii). The system (10) involves the cohorts only through G. The price process is a function of the history of flows, and by Theorem 3.1(i) it depends on the cohorts only through the kernel $k_m = \int r^m \, dM$. A finite measure on $[0,1]$ is determined by G off its support, and also by its moments.

Part (iv). Write $\Phi(\kappa, D; M) = \big(D + \kappa G(D) - 1,\ \kappa(D-\bar{a}) - A(\sigma_F^2 D + \kappa^2\sigma_\xi^2/D)\big)$. At $M = 0$ the zero is $(\kappa_-, 1)$. There the Jacobian with respect to (κ, D) is triangular with determinant $\sqrt{\Delta} \neq 0$: $\partial_D\Phi_1 = 1$, $\partial_\kappa\Phi_1 = G = 0$, and $\partial_\kappa\Phi_2 = (1-\bar{a}) - 2A\kappa_-\sigma_\xi^2 = \sqrt{\Delta}$. The implicit function theorem, applied in the norm of the statement, gives a unique zero near $(\kappa_-, 1)$ that is continuous in M.

Linearizing, $dD = -\kappa_- G(1)$. Differentiating the second equation and using $A(\sigma_F^2 + \kappa_-^2\sigma_\xi^2) = \kappa_-(1-\bar{a})$ gives $d\kappa = -\kappa_-(\bar{a} + 2A\kappa_-\sigma_\xi^2)\, dD/\sqrt{\Delta}$, and so the stated formula. Finally, $G(1) = \sum_i w_i/(1-r_i) = \sum_i q_i$.

At $M = 0$ the transition matrix has eigenvalues $\bar{a}$ and the r_i, all inside the unit disk, so the dynamics remain stable for small M.

Proof of Theorem 4.2. *Part (i).* Write the implicit form of the price at t, $Dx_t = -\kappa H_t + \sum_i \beta_i r_i u_{i,t}$, and the price law at $t-1$, $x_{t-1} = -\kappa H_{t-1} + \sum_i \beta_i u_{i,t}$. Use $H_t = \bar{a}H_{t-1} - \sum_i u_{i,t} - \xi_t$. Then

$$Dx_t - \bar{a}x_{t-1} = \kappa\xi_t + \sum_i (\kappa + \beta_i(r_i - \bar{a}))u_{i,t}.$$

Since $\kappa = \beta_i(D - r_i)$, the coefficient is $\beta_i(D-\bar{a}) = \kappa(D-\bar{a})/(D-r_i)$. Divide by D.

The scaled flows inherit the cohort recursion, because the scaling is constant (`p1_eq_scaled_cohort`). The states $(H_t, u_{\cdot,t+1})$ and $(x_t, \tilde{u}_{\cdot,t+1})$ are related by an invertible linear map, since $\kappa \neq 0$ and every scale factor is nonzero. So the two dynamics are conjugate and have the same eigenvalues.

Part (ii). The effective weights are nonzero because $D \neq \bar{a}$. Apply Theorem 5.3(i) to the closed loop of part (i).

Part (iii). Since $G(D) = \sum_i w_i/(D-r_i) = \sum_i \tilde{w}_i/(D-\bar{a})$, the equation $D = 1 - \kappa G(D)$ reads $D = 1 - \kappa^* D\, \tilde{M}([0,1])/(D(1-a^*))$. Solve for D; then $\bar{a} = a^* D$ and $\kappa = \kappa^* D$. Conversely, $(\bar{a}, \kappa, D, M)$ determine the effective objects. So two equilibria give the same price response if and only if their effective objects agree, and then their merged measures agree. This is consistent with Theorem 3.1.

Part (iv).

- *D exceeds every rate.* The hypothesis of Theorem 5.1 is $a^* + \kappa^* W/(1 - r_c - \Delta\eta) < 1$. It gives $\kappa^* W/(1-a^*) < 1 - r_c - \Delta\eta$, so $D > r_c + \Delta\eta$, which is at least every rate.
- *Each economy is an equilibrium.* With the stated choices, $D + \kappa G(D) = 1$ holds (`p1_eq_realize_D`), and so does $\kappa(D-\bar{a}) = AVD$ with $V = \sigma_F^2 + \kappa^{*2}\sigma_\xi^2$ (`p1_eq_realize_dealer`). By part (i) the equilibrium price response is the given closed loop, which is stable. So each economy is an equilibrium by Theorem 4.1(i).
- *The two economies share everything but the cohorts.* They have the same total effective weight W, so they share D, κ and the primitives, and their price responses are those of Theorem 5.1.
- *Upper bound.* On a class where the effective parameters are known or estimated at the parametric rate (Corollary S1.2), Theorem 5.2 recovers $\tilde{w}$. Then D follows from part (iii), and $w_i = \tilde{w}_i(D-r_i)/(D-\bar{a})$ is Lipschitz on the class.

Proof of Theorem S2.2. *Part (i).*

- *D lies in $(m, 1)$.* Let $\kappa > 0$ and $D > 0$. The dealer equation $\kappa(D - \bar{a}) = AVD > 0$ gives $D > \bar{a}$. The cohort equation is $(D-1)(D-r) + \kappa w = 0$. If $D < r$, it forces $D > 1$, which is impossible; and $D = r$ is excluded. So $D \in (r, 1)$, and $\kappa = \kappa_w(D)$.
- *The equilibria are the solutions of $\Psi_w(D) = A$.* Multiplying the dealer equation by D gives $\kappa(D - \bar{a})D = A(\sigma_F^2 D^2 + \kappa^2 \sigma_\xi^2)$, which is $\Psi_w(D) = A$. Conversely, every such D solves the system.
- *Stability.* By Theorem 4.2(i) the dynamics are those of the closed loop with $c = \kappa^* \tilde{w} = (1 - D)(D - \bar{a})/D \in (0, 1)$. Its characteristic polynomial is $p(\lambda) = \lambda^2 - (a^* + r + c)\lambda + a^* r$. The roots are real and nonnegative, because the discriminant is at least $(a^* - r)^2$ and the coefficients alternate in sign. Next, $p(1) = (1 - a^*)(1 - r) - c = (D - \bar{a})(D - r)/D > 0$, so 1 lies outside the interval between the roots. Their product $a^* r$ is below 1, so both roots are below 1.
- *Existence and count.* Ψ_w is continuous and positive on $(m, 1)$ and tends to 0 at both ends: as $D \to 1$, $\kappa_w \to 0$; as $D \to m$, either $D - \bar{a} \to 0$ or $\kappa_w \to 0$. So its maximum A^* is attained. Every $A \in (0, A^*)$ is taken at least twice, once on each side of a maximizer, and no $A > A^*$ is taken.

Part (ii).

- *A parametrization by κ.* For fixed $\kappa \in (0, (1 - r)^2/(4w)]$, the solutions of $\kappa_w(D) = \kappa$ are $D_\pm(\kappa, w) = \big((1 + r) \pm \sqrt{(1 - r)^2 - 4\kappa w}\big)/2$.
- *The larger root gives the maximum.* $\tilde{\Psi}(\kappa, D) = \kappa(D - \bar{a})D/(\sigma_F^2 D^2 + \kappa^2 \sigma_\xi^2)$ is strictly increasing in D for $D > \bar{a}$. The numerator of its D-derivative is $\bar{a}\sigma_F^2 D^2 + (2D - \bar{a})\kappa^2 \sigma_\xi^2 > 0$. So $A^*(w) = \max_\kappa \tilde{\Psi}(\kappa, D_+(\kappa, w))$.
- *Strict decrease.* Let $w_1 < w_2$, and let κ° attain $A^*(w_2)$. Then κ° is feasible at w_1, and $D_+(\kappa^\circ, w_1) > D_+(\kappa^\circ, w_2)$. So $A^*(w_1) \geq \tilde{\Psi}(\kappa^\circ, D_+(\kappa^\circ, w_1)) > A^*(w_2)$.
- *The limit as $w \downarrow 0$.* The bound $A^* < A_0^*$ of part (iii) is one half. For the other, fix $\kappa_0 = \sigma_F/\sigma_\xi$. As $w \downarrow 0$, $D_+(\kappa_0, w) \to 1$, and $\tilde{\Psi}(\kappa_0, D_+) \to (1 - \bar{a})/(2\sigma_F \sigma_\xi)$.

Part (iii).

- *The bound A_0^*.* By the arithmetic–geometric mean inequality, $\sigma_F^2 D^2 + \kappa^2 \sigma_\xi^2 \geq 2\sigma_F \sigma_\xi \kappa D$. So $\Psi_w(D) \leq (D - \bar{a})/(2\sigma_F \sigma_\xi) < A_0^*$. Machine-checked in squared form as $4A^2 \sigma_F^2 \sigma_\xi^2 \leq (D - \bar{a})^2$ (`p1_fragility_benchmark_bound`).
- *The bound $\Phi^*/(\sigma_F^2 w)$.* Dropping $\kappa^2 \sigma_\xi^2 > 0$ from the denominator gives $\Psi_w(D) < (1-D)(D-r)(D - \bar{a})/(\sigma_F^2 w D) \leq \Phi^*/(\sigma_F^2 w)$. Machine-checked as $A\sigma_F^2 w D \leq (1 - D)(D - r)(D - \bar{a})$ (`p1_fragility_feedback_bound`).
- *The bound on Φ^*.* It uses $(1 - D)(D - r) \leq (1 - r)^2/4$ and $(D - \bar{a})/D \leq 1 - \bar{a}$.
- *The limit as $w \to \infty$.* We have $\kappa_w = O(1/w)$ uniformly on $[m, 1]$. So $w\Psi_w(D) \to (1 - D)(D - r)(D - \bar{a})/(\sigma_F^2 D)$ uniformly, and the maxima converge.

Part (iv).

- *κ is determined by D.* Let $D > \max_i r_i$. All weights are positive, so $G(D) > 0$. Then $D = 1 - \kappa G(D)$ with $\kappa > 0$ gives $D < 1$ and $\kappa = (1 - D)/G(D)$. The dealer equation then reads as in part (i), and $D > \bar{a}$ as there.
- *Comparison with $G(1)$.* For $D \in (\max_i r_i, 1)$, each $D - r_i < 1 - r_i$, so $G(D) > G(1)$.

- *The bound.* The dealer equation gives $A\sigma_F^2 D < \kappa(D-\bar{a}) < (1-D)(D-\bar{a})/G(1)$. Moreover

$$(1-D)(D-\bar{a}) = D(1-\sqrt{\bar{a}})^2 - (D-\sqrt{\bar{a}})^2 \le D(1-\sqrt{\bar{a}})^2.$$

  Machine-checked as `p1_fragility_K_bound`. The bound A_0^* follows as in part (iii).

- *Stability.* By Theorem 4.2(i), the dynamics are those of the closed loop with $(a^*, \kappa^*, \tilde{w})$. Here $a^* \ge 0$, $\kappa^* > 0$, $\tilde{w}_i > 0$ and $r_i \ge 0$, so its transition matrix in the state $(x_t, y_{1,t}, \dots, y_{K,t})$ is nonnegative.
  - *The spectral radius is the root above the rates.* By Perron–Frobenius, the spectral radius ρ is an eigenvalue. Every eigenvalue $z \notin \{0, r_i\}$ solves $F(z) = 1$, where $F(z) = a^*/z + \kappa^* \sum_i \tilde{w}_i/(z - r_i)$. No r_i is an eigenvalue, because $\tilde{w}_i \neq 0$ (Theorem 5.3(i)). F decreases strictly from $+\infty$ to 0 on $(\max_i r_i, \infty)$, so it has exactly one root z^* there, and $\rho \ge z^*$. Being real and at least z^*, ρ equals z^*.
  - *The criterion.* Hence $\rho < 1$ if and only if $F(1) < 1$, that is, $a^* + \kappa^* \sum_i \tilde{w}_i/(1-r_i) < 1$. With $g_i = w_i/(D - r_i)$ this reads $\kappa \sum_i g_i/(1-r_i) < 1$ (`p1_eq_effective_stability_iff`).
  - *It holds.* Each $1 - r_i > 1 - D > 0$ and $\kappa \sum_i g_i = 1 - D$, so $\kappa \sum_i g_i/(1-r_i) < 1$ (`p1_eq_stability_K`).

Proof of Corollary S2.1. *Part (i).*

- The cohort recursion is part of the model.
- By Theorem 4.2(i), the price is the closed loop with $a^* \ge 0$, $\kappa^* > 0$ and $\tilde{w}_i > 0$, driven by the i.i.d. flow ξ.
- It is stable by Theorem S2.2(iv).
- These are the hypotheses of Proposition 6.2. Its regression involves only cohort i's own flow and the price, so it delivers the true (r_i, w_i).

Part (ii).

- *Price-only side.* By Theorem 4.2, the price-only experiment is the reduced form at the effective parameters. The realization of Theorem 4.2(iv) puts the Gauss and Gauss–Radau pair inside the equilibrium class, so the price-only side of Theorem 6.3 holds with the effective weights.
- *Tagged side.* Each cohort's flow response to a unit impulse is $f_{i,t} = w_i \sum_{s \le t} r_i^{t-s} h_s$, where h is the equilibrium response and $h_0 = \kappa^* > 0$. That is the tagged record of Theorem 6.3, and its argument uses only that h is bounded with $h_0 > 0$.
- On the stated classes the true weights are a Lipschitz function of the effective ones, and conversely, so the orders are unchanged.

Numerical checks of Theorems 4.2 and S2.2. The script `equilibrium/e4_reduced_form.py` runs three checks.

- *Reduced form.* For four solved equilibria, the equilibrium price path and the reduced-form path with $(a^*, \kappa^*, \tilde{w})$ agree to 4.4×10^{-16} over 5,000 simulated periods. The four are: one cohort with positive feedback, one with negative feedback, two cohorts, and the Gauss pair. Their impulse responses agree to 10^{-16}. For one cohort of weight 0.02 at $r = 0.7$, for example, $a^* = 0.629$, $\kappa^* = 0.621$ and $\tilde{w} = 0.0279$.

- *Realization.* Take the Gauss and Gauss–Radau pair with $K = 2$, $a^* = 0.5$, $\kappa^* = 0.5$, $W = 0.2$, $r_c = 0.6$ and $\Delta\eta = 0.1$, so that the stability index of Theorem 5.1 is 0.833. Both economies are equilibria with $D = 0.8$, $\bar{a} = 0.4$, $\kappa = 0.4$ and $A = 0.188$, and both equilibrium equations hold to machine precision.
  - Their true cohort weights are $(0.064, 0.036)$ and $(0.0375, 0.0625)$.
  - Their equilibrium impulse responses differ in ℓ_2 by $1.22\,d^3$, $1.25\,d^3$ and $1.29\,d^3$ at $d = 0.1$, 0.05 and 0.025. That is the order $(\Delta\eta)^{2K-1}$.
- *Fragility.* The maximum of Ψ_w reproduces the bisection thresholds of the solver to 10^{-12} ($A^* = 0.2547$, 0.1297, 0.0677, 0.0342 at $w = 0.02$, 0.05, 0.1, 0.2).
  - $A^* = 0.3967$ at $w = 5 \times 10^{-4}$, against the limit 0.4.
  - $w\,A^* = 0.006875$ at $w = 1$ and 0.0068764 at $w = 10$, against $\Phi^* = 0.0068764$.

Several cohorts. `equilibrium/e4_fragility_K.py` scans D for three populations at weights 0.005 to 1 per cohort. The populations are two cohorts at rates 0.5 and 0.85, three at 0.3, 0.6 and 0.8, and the Gauss pair.

- The exact threshold A^* lies below the part (iv) bound in all 15 cases, at 28% to 86% of it. For example, for two cohorts with $G(1) = 1.73$, $A^* = 0.0084$ against a bound of 0.0293.
- Every equilibrium on the threshold curve is stable. At the threshold the spectral radius is between 0.79 and 0.93; it approaches 1 only at the end of the curve where $D \to \max_i r_i$ and $A \to 0$.

The result files are `equilibrium/e4_reduced_form_results.json` and `equilibrium/e4_fragility_K_results.json`.

Proof of Theorem S2.3. *Part (i).*

- *Dealers' information.* The cohorts' flow is predictable. So the dealers' information about θ_t is ν_t, and $E[\theta_t \mid \nu_t] = \lambda\nu_t$ with $\lambda = \mathrm{Cov}(\theta, \nu)/\mathrm{Var}(\nu) = b\sigma_\theta^2/\sigma_\nu^2$.
- *The price decomposes.* $E_t F_{t+1} = F_t + \lambda\nu_t$ and $E_t\nu_{t+1} = 0$, so $E_t p_{t+1} - p_t = E_t x_{t+1} - x_t$. The dealers' condition is therefore the one of Theorem 4.1. Its state is $(H_t, u_{\cdot,t+1})$, and $E_t u_{i,t+2} = r_i u_{i,t+1} + w_i E_t x_{t+1}$. Coefficient matching gives the same equations for β, D and κ (`p1_inventory_necessity`).
- *The price's response to ν_t.* Since the cohorts trade on $s_t = x_t + \lambda\nu_t$ and $\sum_i \beta_i w_i = 1 - D$, the implicit form is $Dx_t = -\kappa H_t + \sum_i \beta_i r_i u_{i,t} + (1 - D)\lambda\nu_t$. The response of s_t to ν_t is therefore $(\kappa + (1 - D)\lambda)/D + \lambda = (\kappa + \lambda)/D$.
- *The insider.* $F_{t+1} - p_t = \theta_t + \varepsilon_{t+1} - m_t - c\,\xi_t - c\,d_t$, where $c = (\kappa + \lambda)/D$ and m_t is the predictable part of s_t. The insider does not observe the public state, and m_t has mean zero and is independent of θ_t. So she maximizes $(\theta - c\,d)\,d$. So $d = \theta/(2c)$ and the profit is $\theta^2/(4c)$ (`p1_insider_best_response`). Taking expectations, $b\sigma_\theta^2/2$.
- *The variance.* The innovation of p_{t+1} given dealers' information is $(\theta_t - \lambda\nu_t) + \varepsilon_{t+1} + c\,\nu_{t+1}$, which gives V (`p1_insider_posterior_variance`).
- *Converse.* A solution of the system satisfies every condition just used.

Part (ii). With $A = 0$, $\kappa(D - \bar{a}) = 0$ forces $\kappa = 0$, and then $D = 1$. Then $b = 1/(2\lambda)$ and $\lambda\sigma_\nu^2 = b\sigma_\theta^2$ give $4\lambda^2\sigma_\xi^2 = \sigma_\theta^2$ (`p1_insider_kyle_limit`).

Part (iii). The cohorts enter only through $D = 1 - \kappa G(D)$.

Part (iv).

- *The reduced form.* Combine the implicit form at t, the price law at $t-1$ and $H_t = \bar{a}H_{t-1} - \sum_i u_{i,t} - \nu_t$, as in Theorem 4.2. This gives $Ds_t - \bar{a}s_{t-1} = \sum_i \beta_i(D-\bar{a})u_{i,t} + (\kappa+\lambda)\nu_t - \bar{a}\lambda\nu_{t-1}$ (`p1_insider_reduced_form`).
- *The filter is invertible.* $c_1 < c_0$ because $\bar{a} < 1$, $\lambda \geq 0$ and $\kappa \geq 0$. So the observer, knowing the primitives, inverts the filter, and is left with the closed loop of Theorem 4.2.
- *Realization.* Proceed as in Theorem 4.2(iv):
  - set $D = 1 - \kappa^* W/(1-a^*)$, $\bar{a} = a^* D$ and $\kappa = \kappa^* D$;
  - solve the insider's pair (λ, b) given (κ, D);
  - set $A = \kappa(D-\bar{a})/(VD)$.

  The two economies then share every primitive.

Numerical check of Theorem S2.3. The script `equilibrium/e6_insider.py` solves the system at $A = 0.05$, $\sigma_\varepsilon^2 = 1$, $\sigma_\theta^2 = \sigma_\xi^2 = 0.25$ and $\bar{a} = 0.6$, and simulates 2×10^5 dates for each solution.

Cohorts	κ	D	λ	b	$(\kappa+\lambda)/D$	profit	inform.
none	0.1656	1	0.4835	0.7704	0.6490	0.0963	0.372
+0.02 at 0.7	0.1692	0.9883	0.4816	0.7592	0.6586	0.0949	0.366
−0.2 at 0.7	0.1463	1.0775	0.4928	0.8430	0.5931	0.1054	0.415
two cohorts, or split†	0.1679	0.9924	0.4823	0.7632	0.6552	0.0954	0.368

Informativeness (inform.) is $b^2\sigma_\theta^2/\sigma_\nu^2$, the share of $\mathrm{Var}(\theta)$ resolved by the price. †Weights $0.005, 0.005$ at rates $0.5, 0.85$, or the same merged measure split into three cohorts; both give identical coefficients.

- In every case, the dealers' first-order residual is below 5×10^{-17}, and the observer's reduced form holds to 4.4×10^{-16}.
- The regression of θ on ν reproduces λ to the third decimal.
- The simulated profit at $b' = 0.6b, \ldots, 1.4b$ peaks at b, as does the analytic profit.
- As $A \downarrow 0$ (from 0.05 to 0.001), (λ, b) moves from $(0.482, 0.759)$ to $(0.49999, 0.994)$, toward Kyle's $(0.5, 1)$.

Numerical solution of Theorem 4.1. The script `equilibrium/e4_inventory.py` solves the system (10) at $A = 0.2$, $\sigma_F^2 = 1$, $\sigma_\xi^2 = 0.25$ and $\bar{a} = 0.6$.

How the solver finds equilibria.

- For one cohort with $w > 0$, every equilibrium with $\kappa > 0$ and $D > 0$ has $D \in (\max(\bar{a}, r), 1)$ and $\kappa = (1-D)(D-r)/w$. A scan over D therefore finds all of them.
- For several cohorts, the solver follows the branch of $D(\kappa)$ that is continuous with $D = 1$.
- Each solution is checked by simulating 2×10^5 periods of the closed loop. The check computes the dealers' first-order residual and the innovation variance of the price.

Benchmark. Here $\Delta = 0.12$, and the two equilibria are $\kappa_- = 0.536$ and $\kappa_+ = 7.46$, as in part (ii).

Results.

Cohorts (weights at memories)	κ	D	κ/D	spectral radius
none	0.5359	1	0.5359	0.600
0.02 at 0.7	0.5916	0.9533	0.6206	0.787
-0.2 at 0.7	0.4212	1.1767	0.3579	0.597
$0.005, 0.005$ at $0.5, 0.85$	0.569147	0.970292	0.586573	0.881
$0.0025, 0.0025, 0.005$ at $0.5, 0.5, 0.85$	0.569147	0.970292	0.586573	0.881
Gauss pair, $0.01, 0.01$	0.5759	0.9649	0.5969	0.766
Radau pair, $0.005, 0.015$	0.5743	0.9662	0.5945	0.738
0.2 at 0.7; or $0.1, 0.1$ at $0.5, 0.85$		no equilibrium		

- *The simulation checks hold in every case.* The relative first-order residual is below 4×10^{-16}. The simulated innovation variance matches $\kappa^2 \sigma_\xi^2 / D^2$ to within 0.3%.
- *Invariance (part (iii)).* The split configuration has the same merged measure as the two-cohort one, and its coefficients agree to all printed digits.
- *Gauss and Radau pairs.* They agree on the first three kernel moments, and their impacts differ by 0.0016.
- *Negative feedback* lowers impact.

First-order formula of part (iv), for one cohort at $r = 0.7$:

- at $w = 0.005$, $\kappa = 0.5457$ against the first-order value 0.5449, and $D = 0.9906$ against 0.9911;
- at $w = 0.01$, $\kappa = 0.5574$ against 0.5540.

Doubling w multiplies the error by about 4.5, as a second-order remainder should.

Existence region. For one cohort at $r = 0.7$, an equilibrium exists for dealer risk aversion below a threshold A^*. This is computed by bisection, with the full scan over D at each step:

w	0	0.02	0.05	0.1	0.2
$G(1) = w/(1-r)$	0	0.067	0.167	0.333	0.667
A^*	0.400	0.255	0.130	0.068	0.034

- Without feedback, $A^* = (1 - \bar{a})/(2\sigma_F \sigma_\xi) = 0.4$, which is the closed form of part (ii).
- For larger $G(1)$, A^* falls roughly in proportion to $1/G(1)$.
- At the threshold the two equilibria merge and disappear.

The results file is `equilibrium/e4_inventory_results.json`.

Numerical checks.

- `equilibrium/e2_rents.py`, with $T = 10$, $\lambda = 0.5$ and $\Delta k_m = \pm 0.4\,\epsilon\,(0.7)^m$:
  - the exact rent is -0.322ϵ when the dealer understates feedback and $+0.322\epsilon$ when it overstates it, as $\epsilon \to 0$;
  - this matches the first-order formula $\mp \Sigma_0 \sum_m \omega_m |\Delta k_m|$ to three digits at $\epsilon = 0.0125$, and to within 6% at $\epsilon = 0.4$.
- `equilibrium/e3_halt.py`, with the market of `e1_demo.py`, $c = 0.6$, $W = 0.5$, 200 draws and 40 dates:
  - the largest pre-halt price gap is 3.8×10^{-4}, 2.9×10^{-3} and 2.3×10^{-2} for $d = 0.05$, 0.1 and 0.2, a ratio to d^3 of 3.0, 2.9 and 2.8;
  - the largest post-halt gap is 0.196, 0.190 and 0.177.

The result files are `e2_rents_results.json` and `e3_halt_results.json`. Parts of both theorems are machine-checked.

- *Theorem S2.4(i):* the symmetrization, the optimal schedule and its value (`Paper1EquilibriumExtras.lean`).
- *Theorem S2.5(i):* the kernel bound $|k_m^G - k_m^R| \le 2W\binom{m}{3}|d|^3(|c|+|d|)^{m-3}$ for every $m \ge 3$ (`Paper1HaltKernelBound.lean`).
- *Theorem S2.5(iii):* the post-halt gap $W/4$.

The remaining parts are proved on paper.

S3 Execution cost and capital

Theorem S3.1 (Sharp probing-resource orders for a known symmetric pair)**.** *Fix $0 \le a < 1$, $\kappa, q > 0$, $S = a + \kappa q < 1$ and $\eta \in (0,1)$. Compare a single cohort of weight q and rate η with two cohorts of weight $q/2$ and rates $\eta \pm \delta$, both in $[\epsilon, 1-\epsilon]$. All parameters are known. The price receives executed quantity through κv_t, starts from zero, has no process noise, and is observed with independent $N(0,\sigma^2)$ noise. Probes are deterministic and common to both alternatives. Let $d_T = (h_j^\delta - h_j^0)_{j<T}$ and $K_* = 2\kappa^2 q/[\epsilon^3(1-S)^2]$.*

(i) For every probe with $\sum_t |v_t| \le V$,

$$D_{\mathrm{KL}}(P_0^v \| P_\delta^v) \le \frac{K_*^2 V^2 \delta^4}{2\sigma^2}, \qquad e_{\mathrm{worst}} \ge \frac{1}{2}\Big(1 - \frac{K_* V \delta^2}{2\sigma}\Big)_+. \tag{13}$$

(ii) The optimal worst-case error over these probes and all classifiers is exactly

$$e_T^*(V) = \Phi\left(-\frac{V\|d_T\|_2}{2\sigma}\right), \tag{14}$$

attained by a pulse at the first input and the likelihood-ratio test. Waiting adds information but $\|d_T\|_2 \uparrow \|d_\infty\|_2 < \infty$, so it cannot make the error arbitrarily small at fixed V. Error at most $\alpha < 1/2$ requires $V \ge 2\sigma(1-2\alpha)/(K_*\delta^2)$ independently of T.

(iii) The first two taps coincide and $d_2 = -\kappa^2 q\delta^2$. A single pulse of size

$$A = \frac{2\sigma\Phi^{-1}(1-\alpha)}{\kappa^2 q \delta^2} \tag{15}$$

and the third observation achieve error exactly α. The necessary and sufficient quantity order is δ^{-2}.

(iv) Under average power $\sum_{t<T} v_t^2 \le PT$, the necessary horizon order is δ^{-4}, attained by the pulse whenever $T \ge \max\{3, \lceil A^2/P \rceil\}$.

(v) For a fixed monetary price exposure $\lambda \ne 0$, additive reserve L and cost envelope $c\sum|v_t|$, the sharp robust quantity allowance under budget $B > L$ is $V_B = (B-L)/(|\lambda|\kappa + c)$.

In words. To tell one desk from two desks whose memories differ slightly, the observer must trade against them, and the quantity it must trade grows like $1/\delta^2$: halving the difference quadruples the

trading needed. Waiting longer after a fixed trade does not help, because the trade's echo in the price dies out.

A probe changes inventory as well as price. We therefore specify cash and execution timing. Price has units of currency per asset, and κ currency per squared quantity.

Lemma S3.2 (Fair execution)**.** *For a quote schedule refreshed after the uncontrolled state update, execution at the average price of a linear book costs, relative to the constant fundamental mark,*

$$C(v) = \tfrac{1}{2} v^\top T_h v + c\|v\|_1, \qquad (T_h)_{ij} = h_{|i-j|}. \tag{16}$$

In the positive stable economy T_h is positive definite, so no nonzero round trip produces a deterministic gain.

Positivity here means that the response is a positive mixture of decaying exponentials (Lemma S1.1), whose spectral density is positive. Positive, decreasing taps alone do not suffice: for $h = (1, 0.9, 0.1)$ the 3×3 matrix T_h has determinant -0.468, so some round trip earns money. The conclusion concerns modelled execution cost only.

We verify the ledger identity (16) against an explicit cash account to 5.7×10^{-13} on 1,000 random probes, and the smallest eigenvalue of T_h over 1,000 random stable positive economies is 0.141. A direct-only proxy $\sum \frac{1}{2}\kappa v_t^2$ omits cross-time impact: for $v = (+1, -1)$ it gives 0.80 against a true Toeplitz loss of $h_0 - h_1 = 0.3488$.

S3.1 Capital supplied by a counterparty

Theorem S3.3 (Sharp robust quantity cap from cash and margin)**.** *Use the positive, stable, reset economy with constant fundamental price $P_* > 0$ and no process noise. A designated dealer fills only nonnegative probes $v_t \geq 0$ at the average price $P^0_{t+1} + \kappa v_t/2$ of a linear schedule, holds inventory $I_{t+1} = I_t - v_t$ and cash $K_{t+1} = K_t + v_t(P^0_{t+1} + \kappa v_t/2) - cv_t$, starts with equity $B > 0$, and must satisfy $\mu|I_t|P_t \leq E_t$ at every time. For the robust contract admitting every probe with $\sum_t v_t \leq V$, the largest allowance guaranteeing margin is*

$$V_B = \frac{2B}{\beta + \sqrt{\beta^2 + 4(\mu + \frac{1}{2})\kappa B}}, \qquad \beta = c + \mu P_*, \tag{17}$$

equivalently $(\mu + \frac{1}{2})\kappa V^2 + (c + \mu P_*)V \leq B$. The cap holds across all positive cohort representations with $S < 1$, without resolving individual rates. An observer that receives only noisy prices and executed quantities has best error $\Phi[-V_B\|d_T\|_2/(2\sigma)]$ within this allowance; fixed error $\alpha < 1/2$ requires capital of order at least δ^{-4}, and a single positive pulse attains that order.

In words. A probe moves the price against the account that makes it, so the margin it needs grows with the square of the probe. Because the probe grows like $1/\delta^2$, the capital grows like $1/\delta^4$: halving the difference multiplies the capital needed by 16.

Even with zero initial inventory, the probe creates a quadratic adverse mark that a fixed initial price exposure would miss. Figure S1 illustrates the experiment.

Proposition S3.4 (Exact financial disclosure changes the experiment)**.** *In this noiseless-state account, exact disclosure of the third equity mark after a single positive pulse $0 < A \leq V_B$ distinguishes the known pair with zero error for every $\delta > 0$.*

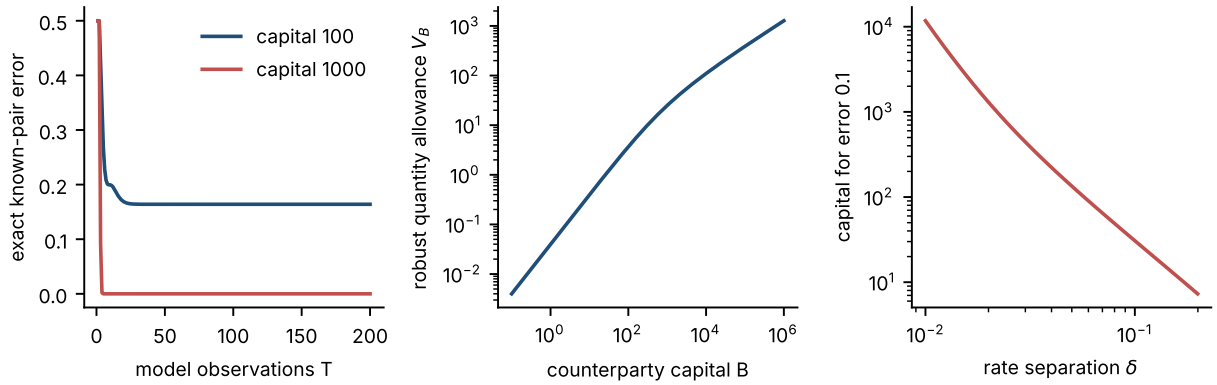


Figure S1: Known-pair experiment funded by the counterparty cash/margin contract. (a) Exact Gaussian error at fixed capital 100 and 1,000, separation $\delta = 0.05$. (b) The sharp robust quantity allowance (17). (c) Capital for error 0.1 with 200 observations. Parameters: $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.4$, observation s.d. 0.003, reference price 100, margin 0.25, cost 0.02. No process noise.

In words. If the observer can see an account's exact equity, a single test trade settles the question, however close the two candidate markets are. The hard part of the problem is therefore not the market itself but what the observer is allowed to see.

Exact marked equity is extra information, not a free observation assumed by the price-only theorem. Real account marks have their own precision: with reports rounded to 0.01, a guaranteed change requires a larger trade (see the disclosure contracts of the full working paper).

S3.2 Capital paid by the probe trader

Lemma S3.5 (Time-domain Toeplitz coercivity: capital order for funded probes)**.** *Use the same economy and execution law. The statistical observer receives only noisy prices and executed quantities. The probe trader starts with cash $B > 0$ and zero inventory; a finite signed sequence v changes inventory by $+v_t$ and cash by $K_{t+1} = K_t - v_t(P^0_{t+1} + \kappa v_t/2) - c|v_t|$, after which trading stops and the trader must remain solvent as the price returns to P_*. Put*

$$r = 1 - \epsilon(1 - S) < 1, \qquad m = \kappa \frac{1 - r}{1 + r} > 0, \qquad K_* = \frac{2\kappa^2 q}{\epsilon^3 (1 - S)^2}.$$

Every solvent strategy satisfies $D_{\mathrm{KL}}(P^v_0 \| P^v_\delta) \le K_*^2 B \delta^4/(\sigma^2 m)$, so error at most $\alpha < 1/2$ requires

$$B \;\ge\; \frac{2\sigma^2 m (1 - 2\alpha)^2}{K_*^2 \delta^4}, \tag{18}$$

without a quantity cap and at any horizon. Conversely, for any margin fraction $0 < \mu \le 1$, the pulse A in (15) achieves error α and remains margin-solvent at every later mark with capital $B_A = \frac{\kappa}{2} A^2 + (c + \mu P_*) A$. Necessary and sufficient capital orders are δ^{-4}.

Lemma S3.5 is an order statement: an elementary time-domain coercivity argument for Toeplitz forms that establishes the δ^{-4} rate before the exact frequency-domain optimization of the account-level analysis of the full working paper. Its explicit constant is very conservative, because it scales

as $m/K_*^2 \propto \epsilon^7(1-S)^5$: at the calibrated parameters it is 4.9×10^{-4} par units at $\delta = 0.05$ with $\epsilon = 0.35$, five orders of magnitude below any feasible design. We therefore state it as a lemma; the account-level analysis of the full working paper computes the sharp constant, within 5% of a feasible design at every separation.

Theorem S3.1. In either positive economy $J \geq 0$ and $\|J\|_\infty \leq 1-\epsilon(1-S) < 1$, so the response is nonnegative with $\max h = \kappa$ and $\sum h = \kappa/(1-S)$. From the display above, $|G_\delta - G_0| \leq 2q\delta^2/\epsilon^3$ on the circle; combined with the resolvent identity and positivity this gives $\|H_\delta - H_0\|_\infty \leq K_*\delta^2$, and Parseval bounds $\|h^\delta - h^0\|_2$ by the same constant. Young's inequality gives $\|\mu_\delta - \mu_0\|_2 \leq K_*\delta^2\|v\|_1$, and Gaussian KL with Pinsker gives (13). For (ii), the mean difference is a lower-triangular convolution matrix times v; each column is a shifted copy of a prefix of d_T, so $\|\mu_\delta - \mu_0\|_2 \leq V\|d_T\|_2$ with equality for a pulse at $t = 0$. For two known equal-covariance Gaussian laws the minimax error is $\Phi(-\|\mu_\delta - \mu_0\|_2/(2\sigma))$, attained by the midpoint likelihood-ratio test. For (iii), the merged measures have equal mass and their first moments differ by $-q\delta^2$, which enters h_2 with factor κ^2 while h_0, h_1 coincide; a pulse A separates the third-observation means by $A\kappa^2 q\delta^2$. Part (iv) follows from the ℓ_2 gain bound, and (v) from superposition and $\max_j h_j = \kappa$, which give $\sup_t |\lambda x_t| \leq |\lambda|\kappa V$, attained by a single pulse.

Lemma S3.2. After the uncontrolled update, the pre-probe displacement at step j is $\sum_{i<j} h_{j-i}v_i$; average linear-book execution adds $\kappa v_j/2$. Summing signed execution costs gives $\frac{1}{2}v^\top T_h v$. Each kernel $\lambda^{|i-j|}$ is a positive-definite AR(1) covariance for $0 \leq \lambda < 1$, and a nonnegative mixture with positive total weight is positive definite. For a round trip the fundamental cash term cancels.

Theorem S3.3. Let $U_t = \sum_{j<t} v_j$. Positive responses give $0 \leq x_t \leq \kappa U_t$ and $I_t = -U_t$. Summing cash and marking inventory, $B - E_t = U_t x_t - \sum_{j<t} v_j x^0_{j+1} - \frac{\kappa}{2}\sum_{j<t} v_j^2 + cU_t$. Its quadratic part $v^\top A_t v$ has diagonal $h_{t-1-i} - \kappa/2$ and off-diagonal $\frac{1}{2}(h_{t-1-i} + h_{t-1-j} - h_{j-i})$ for $i < j$; monotonicity of h bounds every entry by $\kappa/2$. Hence $B - E_t \leq \kappa U_t^2/2 + cU_t \leq \kappa V^2/2 + cV$, while position value is at most $V(P_* + \kappa V)$; the stated inequality guarantees margin. A single pulse attains both adverse bounds at its first response, so no larger allowance is robust. Solving the quadratic gives (17). The quantity requirement $V_B \gtrsim \delta^{-2}$ then forces $B \gtrsim \delta^{-4}$, and the pulse A funded with $(\mu + \frac{1}{2})\kappa A^2 + (c + \mu P_*)A$ attains it.

Proposition S3.4. The initial cash receipt is common to both alternatives and the dealer then holds $-A$. At time three $E_{3,r} = B + \kappa A^2/2 - cA - A^2 h_{r,2}$, so $E_{3,\delta} - E_{3,0} = A^2\kappa^2 q\delta^2 > 0$.

Lemma S3.5. Marking residual inventory at P_* gives $E_\infty = B - \frac{1}{2}v^\top T_h v - c\sum|v_t| \geq 0$. For each mode $\lambda \leq r$ the Toeplitz kernel has spectral density $(1-\lambda^2)/(1-2\lambda\cos\omega + \lambda^2) \geq (1-r)/(1+r)$, so $v^\top T_h v \geq m\|v\|_2^2$ and $\|v\|_2^2 \leq 2B/m$. The transfer bound gives mean-difference squared norm at most $K_*^2\delta^4\|v\|_2^2$, hence the KL bound; Pinsker gives (18). For one pulse, cash becomes $B_A - A(P_* + \kappa A/2) - cA$ and inventory A; at later marks $E_t = \mu A P_* + A^2 h_{t-1}$ and $\mu|I_t|P_t = \mu A P_* + \mu A^2 h_{t-1}$, so margin holds for $\mu \leq 1$.

S4 Two unknown cohorts: pole fitting and the global two-point bound

Because the memory rates are zeros of H (Theorem 3.3), extracting poles directly from the price impulse response by matrix-pencil or Prony methods targets closed-loop eigenvalues. A three-step inversion repairs this: (1) compute a minimal realization of H itself from the Hankel matrix of $h_1, h_2, \ldots$ (with h_0 as the direct term) by Kung's eigensystem realization; (2) compute its transmission zeros, which are the memory rates and the timing zero at 0 (transforming the sequence first, for example to $h_{t+1} - ah_t$, cancels these zeros); (3) project reconstructed flow onto the recovered memory filters to obtain the weights (full working paper, internet appendix). Without noise this recovers $K = 0$, $K = 1$ and well-separated two-cohort systems exactly; with noise, per-cohort weights become unreliable once rates are close. The reason is information-theoretic, as the rest of this section shows.

S4.1 A global two-point bound for any estimator

A Cramér–Rao benchmark (full working paper) applies only to estimators that are unbiased near the true parameter. Could biased, ridge, shrinkage or Bayesian estimators escape the $(\Delta\eta)^{-6}$ scaling? No.

Theorem S4.1 (Global two-point bound and matching upper bound, $K = 2$)**.** *In this theorem w_k denotes cohort k's trading intensity q_k, not the merged weight $q_k\eta_k$ of Section 2, and $\Delta\eta$ is the actual spacing of the two rates. Fix total intensity $q = 0.20$, mean rate $\bar{\eta} = 0.40$, target separation $\tau = 0.01$, and $\Delta\eta \in (0, 8/11)$ (so that $\bar{\eta} - p_+\Delta\eta > 0$). Let $p_\mp = \frac{1}{2} \mp \tau/q$ and*

$$\theta_0 = (qp_-, qp_+, \bar{\eta} - p_+\Delta\eta, \bar{\eta} + p_-\Delta\eta), \qquad \theta_1 = (qp_+, qp_-, \bar{\eta} - p_-\Delta\eta, \bar{\eta} + p_+\Delta\eta). \tag{19}$$

(i) Label cohorts in increasing order of rate, so that weight vectors are compared rate by rate (as unordered sets the two weight vectors coincide). Both economies have spacing $\Delta\eta$, weight separation $\|w(\theta_1) - w(\theta_0)\|_\infty = 2\tau$ in this labelling, and equal rate moments $M_m = \sum_k w_k\eta_k^m$ for $m \in \{0, 1, 2\}$, while

$$M_3(\theta_1) - M_3(\theta_0) = 4\tau p_- p_+ (\Delta\eta)^3. \tag{20}$$

Hence $\|h(\theta_1) - h(\theta_0)\|_2 = \Theta(\tau(\Delta\eta)^3)$.

(ii) For any measurable estimator (including biased, ridge, James–Stein or MAP) that reports weights in the same rate order, the supremum being over the class of two-cohort economies with total weight q, mean rate $\bar{\eta}$ and spacing $\Delta\eta$,

$$\sup_\theta P_\theta(\|\hat{w} - w(\theta)\|_\infty \geq \tau) \;\geq\; 1 - \Phi\Big(\frac{\sqrt{N}\|h(\theta_1) - h(\theta_0)\|_2}{2\sigma_h}\Big), \tag{21}$$

so failure probability at most 0.05 requires $N \geq 4\sigma_h^2 z_{0.95}^2/\|h(\theta_1) - h(\theta_0)\|_2^2 = \Omega((\Delta\eta)^{-6})$. For $\Delta\eta \geq 2\tau q/(q + 2\tau)$ (≈ 0.018 at the stated values) the same bound holds without any labelling convention, for the permutation-invariant loss

$$d_\infty(\hat{\theta}, \theta) = \min_{\pi \in S_2} \max_k (|\hat{w}_{\pi(k)} - w_k| + |\hat{\eta}_{\pi(k)} - \eta_k|),$$

because then $d_\infty(\theta_0, \theta_1) = \min\{2\tau(1 + \Delta\eta/q), (1 + 2\tau/q)\Delta\eta\} \geq 2\tau$. For smaller spacings the relevant permutation-invariant statement is the Wasserstein bound of Theorem 5.1.

Table S1: Two-point sample complexity for two-cohort weight recovery

$\Delta\eta$	ΔM_3	$\|\Delta h\|_2$	N^*_{LeCam} (95%)	N_{CR}
0.30	2.67×10^{-4}	1.98×10^{-3}	7.1	47.8
0.20	7.92×10^{-5}	5.60×10^{-4}	88.2	597.0
0.14	2.72×10^{-5}	1.89×10^{-4}	775.3	5,267.9
0.10	9.90×10^{-6}	6.84×10^{-5}	5,927.6	40,374.2
0.08	5.07×10^{-6}	3.49×10^{-5}	22,740.5	155,048.9
0.06	2.14×10^{-6}	1.47×10^{-5}	128,333.0	875,732.3
0.05	1.24×10^{-6}	8.50×10^{-6}	383,859.1	2,620,321.2
0.04	6.34×10^{-7}	4.35×10^{-6}	1,466,361.5	10,012,662.4

Worst-case failure at $\Delta\eta = 0.10$, $N = 50$, *1,000 trials per hypothesis*: bound 44.1%; ML 94.5%, ridge 47.5%, positive-part shrinkage 100%, heavy prior 73.9%.

$\tau = 0.01$, $\sigma_h = 1.6 \times 10^{-3}$, $a = 0.5$, $\kappa = 0.8$, 121 taps. Estimators: ML is least squares from four starts; ridge and heavy prior are posterior modes with prior standard deviations $(0.05, 0.10)$ and $(0.005, 0.02)$ for (weights, rates) around the symmetric centre of the two hypotheses; positive-part shrinkage pulls the ML weights toward their mean. Failure means some weight is off by at least τ.

(iii) Conversely, the matching upper bound $N = O((\Delta\eta)^{-6})$ *on rich separated two-cohort classes is Theorem 5.2 with* $K = 2$. *On the separated class with weights and rates bounded away from the boundary, Toeplitz deconvolution followed by a Prony–Hankel root step has Hankel determinant* $g_1 g_3 - g_2^2 = w_1 w_2 \eta_1 \eta_2 r_1 r_2 (\Delta\eta)^2 > 0$ *for the deconvolved kernel* $g_m = \sum_i w_i \eta_i r_i^m$, *and perturbation analysis gives the same order.*

In words. For two unknown groups, no method can tell the two candidate markets apart reliably with fewer than a number of experiments of order $(\Delta\eta)^{-6}$, and a practical method (Prony's method, Theorem 5.2) achieves this order. Halving the spacing multiplies the data needed by 64.

We recomputed the two-point quantities exactly (Table S1): the moment gap equals $4\tau p_- p_+ (\Delta\eta)^3$ to machine precision, and the Le Cam requirement at $\Delta\eta = 0.10$ is 5,927.6 resets. The worst-case failure rates of four re-implemented estimators all lie above the lower bound (Table S1).

Remark S4.2 (The hard directions carry no risk). The two economies of Theorem S4.1 have the same total weight $\sum_i q_i = q$, so their summed price response $\sum_t h_t = \kappa/(1 - a - \kappa \sum_i q_i)$ is identical, 2.353 at the calibration of Table S1: the directions in which prices are least informative about the cohorts are directions in which cumulative price risk does not move. The summed response is a linear functional of the response and is estimated at the parametric rate whatever the spacing of the memories; the companion paper turns this into a certification theorem, a minimax-optimal pre-trade guard and a decision value of disclosure (Niroula et al., 2026). Identification and safety ask different questions of the same data.

S5 The supervisory decision in detail

This section completes Section 6.

Proof of Proposition 6.1.

Proof of Proposition 6.1. Part (i) follows by scaling the intensities. For (ii), expand $(1-r_c-\Delta\eta s)^{-1}$: the first three moments agree, so aggregate budgets differ by order $(\Delta\eta)^3$. Each intensity converges to $W\omega_i/(1-r_c)$, giving the stated dominant-weight gap. This exceeds the aggregate discrepancy at sufficiently small spacing. A ceiling between the two post-halt budgets reduces the action problem to testing two Gaussian mean vectors. Their common-covariance testing error is $\Phi(-\sqrt{N}\|h^G - h^R\|_2/(2\sigma))$; the response bound in Theorem 5.1 gives the lower bound. Part (iii) follows from continuity of $w/(1-r)$ on the declared compact parameter set and the fixed margin. □

General-K scope. The same reduction gives exponent $4K-2$ only for a pair whose price responses have that moment-matching order *and* whose dominant intensities have a separately established fixed gap. Moment matching alone does not supply the latter. In particular, the symmetric odd/even binomial pair has equal maximal normalized weights. The $K=3$ rows below are numerical examples, not a construction proving this condition for every K.

Explicit robustness condition. Smooth best responses alone do not preserve separation: the constant smooth map $\Phi(M)=0$ sends both post-halt budgets to a. Instead assume that, over the stated action and holding horizon, $|\widetilde{R}_j - R_j| \le \varepsilon$ for $j \in \{G,R\}$, with $R_R < R_G$. Then

$$R_R + \varepsilon \le \bar{s} < R_G - \varepsilon \quad \Longrightarrow \quad \widetilde{R}_R \le \bar{s} < \widetilde{R}_G. \tag{22}$$

Indeed, $\widetilde{R}_R \le R_R + \varepsilon$ and $\widetilde{R}_G \ge R_G - \varepsilon$. A nonempty robust window exists if $2\varepsilon < R_G - R_R$. This is a maintained response bound, not an equilibrium result. The same allowance must cover the alternative uniform action. At the $K=2$, $\Delta\eta = 0.10$ calibration, $R_R = 0.532$, $R_G = 0.569909$, and $\bar{s} = 0.55$ has margin 0.018; an allowance 0.005 fits within it.

A complete conditional cost-and-error model. Let j denote the economy, π_j its declared prior, H the holding horizon, b the surplus per active class per period, C_h the implementation cost of halting and L the penalty per period for violating the ceiling. These are assumed accounting primitives, not estimated social welfare. An envelope $S_{\max}$ supports the robust uniform action

$$\theta_\varepsilon = \min\{1, \max\{0, (\bar{s} - \varepsilon - a)/(S_{\max} - a)\}\}, \qquad L_T = HbK(1-\theta_\varepsilon).$$

This action is feasible only if its worst-case budget meets the ceiling (in particular $a + \varepsilon \le \bar{s}$). For a halt,

$$L_{h,j} = Hb + C_h + HL\,\mathbf{1}\{R_j + d_j > \bar{s}\}, \quad \mathcal{L} = F + Nc + \sum_j \pi_j\{p_j L_{h,j} + (1-p_j)L_T\}.$$

Here $F + Nc$ is the information bill, p_j the probability of halting, and $|d_j| \le \varepsilon$ the action response. Require $\Pr_j(\text{halt and violate}) \le \alpha$ separately in each world. Compare this loss with the best feasible no-purchase lottery over the *same two actions*, and allow refusing to buy information. Neither arbitrary alternative policies nor an optimum over sample size are claimed. Active probing additionally incurs its execution loss and capital requirement; these are not automatically the per-record fee c and must not be counted twice.

Existing runs and the remaining synthetic check. The archived `R_P1_targeted_intervention.json` supplies six response/decision rows. Recomputing them independently from the stated state recursion agrees to 2.1×10^{-17} in the response discrepancy. Those rows supply S_j, R_j

and Gaussian testing distances; they contain no measured disclosure fees, intervention payoffs or strategic response bounds. The existing GPU and historical diagnostics cannot fill these quantities by relabelling their outputs. The new CPU study `studies/supervisory_release_20261105/run_study.py` therefore evaluates only this missing accounting layer, with exact Gaussian probabilities rather than a new large simulation.

For this check alone, assume independent scalar disclosures $Z_n \sim N(R_j, \sigma_t^2)$ and halt when $\bar{Z} + z_{1-\alpha}\sigma_t/\sqrt{N} + \varepsilon \le \bar{s}$. This is an illustrative measurement contract, *not* a claim that class positions or tagged flow directly observe post-halt budgets. If a world is unsafe under the response allowance, this gate falsely halts with probability at most α under the stated unbiased Gaussian law. The price comparator observes the full 400-tap Gaussian reset record of the same known pair, with a likelihood-ratio rule constrained by the same per-world error budget. Both information channels have the same assumed monetary fee here; neither represents free passive observation.

With $K = 2$, $N = 32$, $\sigma_t = 0.02$, $\varepsilon = 0.005$, $\alpha = 0.05$, $\pi_R = 1/2$, $H = b = 1$, $C_h = 0.02$, $L = 10$, $F = 0.05$, and $c = 0.0005$, the uniform loss is 1.449219, the information bill is 0.066, and disclosure loss is 1.305132. Its conditional net advantage is 0.144087 surplus units. Halt probabilities are approximately zero in G and 0.978928 in R. Buying the price record yields loss 1.515219 at this sample size; its optimal acquisition decision is to decline and retain the no-data action. An informed oracle before information fees has loss 1.234609.

Across 2,304 deterministic configurations varying fees, noise, horizon, prior, sample size and allowance, paid disclosure improves on the no-data benchmark in 1,207, worsens it in 1,079, and ties in 18. These counts have no population or statistical-significance interpretation. Negative cases remain in the output. A downward measurement bias of 0.03 produces false-halt probability 0.418776; a future response of 0.025 despite an allowance of 0.005 makes halting unsafe in R with probability 0.978928. The latter also invalidates the uniform-action guarantee, so payoff comparisons under that misspecification are not presented as a fair safety comparison. The experiment establishes conditional accounting and failure modes, not real-market gains, a validated equilibrium response, or the general-K theorem.

Proposition S5.1 (A robust scalar disclosure gate)**.** *Let $Z_1, \dots, Z_N$ be independent $N(R, v)$ readings of the fixed post-halt feedback budget, $N > 0$, and let the realized action response d satisfy $d \le \varepsilon$. For $t > 0$, halt only if $\bar{Z} + t + \varepsilon \le \bar{s}$. Then the probability of halting while the realized budget exceeds the ceiling is at most*

$$\Pr\{\bar{Z} + t + \varepsilon \le \bar{s},\ R + d > \bar{s}\} \le \frac{v}{Nt^2}.$$

Consequently $v \le \alpha N t^2$ gives a per-world error bound α.

Proof. On an unsafe-halt event, $\bar{Z} + t + \varepsilon \le \bar{s} < R + d \le R + \varepsilon$, so $t \le |\bar{Z} - R|$. Chebyshev's inequality for the actual Gaussian sample mean gives the bound. Lean checks the product Gaussian law, the sample statistic, this event inclusion and the probability inequality in `p1_supervisory_gaussian_unsafe_halt`; it is not merely a check of a scalar named "probability bound." The response allowance and the interpretation of Z remain explicit assumptions. □

This conservative gate uses $t = \sigma_t/\sqrt{N\alpha}$. It differs from the sharper Gaussian-quantile gate above, whose exact normal probabilities are numerically evaluated but whose quantile identity is not formalized here. Under the same calibration with $N = 128$, the conservative gate has radius 0.007906, information bill 0.114, loss 1.349034 and conditional net advantage 0.100185. Its halt probabilities are approximately zero in G and 0.998023 in R. Across the same 2,304 sensitivity configurations it improves on the fixed-menu no-data benchmark in 781, worsens it in 1,499 and

ties in 24. Both gates' rows remain in the result file: 4,608 configurations in total. The welfare and error interpretation remains conditional on the declared observation and response contracts.

Operational bridge from tagged flows to a halt decision. The scalar-reading contract above can be replaced, within the reset model, by five observed channels: two first class-flow responses, two second responses, and the first price response. For a unit pulse write $f_{i,0} = \kappa w_i$, $f_{i,1} = w_i(\kappa r_i + h_1)$, and $h_1 = \kappa a + \kappa^2 \sum_i w_i$. Thus

$$w_i = f_{i,0}/\kappa, \qquad r_i = f_{i,1}/f_{i,0} - h_1/\kappa, \qquad q_i = w_i/(1 - r_i).$$

Assume known impact, verified stable class membership, $0 \le r_i \le 0.95$, independent unbiased Gaussian reset readings with per-channel variance σ_f^2, and N independent records. A simultaneous radius $e = \sigma_f \sqrt{5/(N\alpha)}$ covers the five sample means with probability at least $1 - \alpha$ by Chebyshev and a union bound. Interval division, only when the first-flow lower endpoint is positive, gives bounds on w_i, r_i, q_i; intersect rate bounds with the declared class. Halt only if the remaining upper budget plus the response allowance meets the ceiling. Otherwise retain the same robust uniform fallback. On simultaneous coverage, every selected halt is safe. This supplies a conditional observation-to-action argument without feeding the unknown R, w or r to the rule.

The separate Lean module `Paper1FlowDecision.lean` covers exact flow inversion, ratio and clipped intervals, the intensity interval, selected-halt inequality, actual product-Gaussian joint mean bound and a probability-composition theorem with an explicit rule-soundness premise. Their deterministic composition, clipping implementation and executable simulation still require separate correspondence checks; no claim of a fully formalized executable decision algorithm is made. The pinned build receipt records the checked declarations.

The exploratory script `studies/paper1_uplift_20261105/flow_decision.py` runs 20,000 independent replications per world in each of 60 configurations: $N \in \{32, 128, 512, 2048, 8192\}$, $\sigma_f \in \{0.0005, 0.002, 0.005\}$, $H \in \{1, 10\}$ and $F \in \{0.05, 0.25\}$. Other quantities use the two-group calibration above, a per-channel-vector record fee 0.0005, and the same class-halt or throttle action menu. Fourteen configurations have positive estimated net information value; the others are retained. The largest observed gain is exploratory, selected from the grid: at $N = 128$, $\sigma_f = 0.0005$, $H = 10$ and $F = 0.05$, no-data loss is 14.492188, information-inclusive loss 12.370094 and net gain 2.122094 surplus units. A simultaneous grid-adjusted Hoeffding interval is reported in the result file; zero observed unsafe halts is not a proof of zero probability. Costs, horizons and response bounds are synthetic, and no real-market gain or calibrated welfare conclusion follows. Six additional fixed stress cases retain both smaller biases with no observed unsafe halt and failures: a downward second-flow bias of 0.010 produces unsafe halts in G in all 20,000 replications, while a response 0.030 exceeding the allowance 0.005 produces unsafe halts in R in all replications. Swapped second-flow labels lead to abstention in both worlds. These are synthetic contract falsifications, not guarantees under misspecification.

Proposition S5.2 (No-data action and information purchase)**.** *For the fixed halt/throttle menu, let* $\overline{L}_h = \sum_j \pi_j L_{h,j}$ *and let* $p_{\max} = \alpha$ *if a halt is unsafe in at least one world, otherwise* $p_{\max} = 1$. *Assuming the fallback meets the ceiling, the best constant no-data halt probability is* $p = 0$ *when* $\overline{L}_h \ge L_T$, *and* $p = p_{\max}$ *when* $\overline{L}_h < L_T$. *A specified information rule with expected action loss* L_I *is strictly preferred if and only if*

$$F + Nc < \mathcal{L}_0 - L_I.$$

Allowing refusal to purchase gives loss $\min\{\mathcal{L}_0, F + Nc + L_I\} \le \mathcal{L}_0$.

Proof. The no-data loss is affine in its feasible halt probability: $L_T + p(\overline{L}_h - L_T)$. Its slope determines the minimizing endpoint. Subtracting the information rule's total loss gives the break-even inequality, and the minimum inequality proves the optional-purchase statement. These algebraic decisions are machine checked in `Paper1SupervisoryDecision.lean`. They do not establish an optimum over other actions, sample sizes, or information contracts. □

S6 Saturating traders: proofs and checks

Theorem 7.1.

Proof. $f_t = \sum_i c_i g_t(r_i, \ell_i)$ with $g_t(r, \ell) = \psi(\ell \sum_{j<t} r^{t-1-j} s_j)$, so equal measures give equal flows for every signal. The rest is the induction of Theorem 3.1(iii), which uses only that the two flow maps agree and depend on signals before t. The last sentence follows from the example below Theorem 7.2. □

Theorem 7.2.

Proof. (i) By strong induction on $t < \tau$. If the paths agree before t, the signals agree before t, so every trader of either population has the same state in both markets. All are inside their bands, so both aggregate flows equal the linear flow $\sum_{j<t} k_{t-1-j} s_j$, which is the same for P and P' because the kernels agree. The insider's order and the dealer's price are causal, so they agree at t. The same argument with P' replaced by the linear market (the band-linear shape $\psi'(u) = u$) gives the second statement. (ii) Build both markets, and the record noise, on one probability space. By (i) the records coincide on $\{\tau \geq T\}$, so the total-variation distance is at most $\Pr(\tau < T)$, and τ is computed on the linear market. (iii) In the linear market from rest, $x = g * w$ with $g = h/\kappa$, and $m_t(r) = (\varphi_r * x)_t$ with $\varphi_r(j) = r^j$ shifted by one date. So $m_t(r)$ is centred Gaussian with variance at most $s_r^2 \leq \sigma_w^2 \|\varphi_r\|_1^2 \|g\|_1^2$. Here $\|\varphi_r\|_1 = 1/(1-r)$. Also $|h_t|$ is dominated by the response of the positive economy with persistence a and kernel $|k_m|$, whose total mass is $\kappa/(1 - S_{\mathrm{abs}})$, so $\|g\|_1 \leq 1/(1 - S_{\mathrm{abs}})$. For a standard normal Z and $z \geq 0$, $\Pr(|Z| > z) \leq e^{-z^2/2}$. A union bound over the T dates and $n + n'$ traders gives the bound. A test with both errors at most α needs $\mathrm{TV} \geq 1 - 2\alpha$. (iv) Without process noise, the linear path is deterministic, so $\tau \geq T$ surely and (ii) gives distance zero. For a pulse, the linear state is $m_t(r) = A \sum_{j<t} r^{t-1-j} h_j$. □

Proposition 7.3.

Proof. For $u \geq 0$, $\phi(u) = u - \tanh u$ has $\phi(0) = 0$ and $\phi'(u) = \tanh^2 u \in [0, u^2]$. So $0 \leq u - \tanh u \leq u^3/3$, and by oddness $|\tanh u| \leq |u|$ and $|\tanh u - u| \leq |u|^3/3$ for all u. (i) *Size of each path.* Since $|\tanh u| \leq |u|$, $|x_t| \leq a|x_{t-1}| + \kappa \sum_i c_i |\ell_i| \, (\varphi_{r_i} * |x|)_t$ for $t \geq 1$, with $|x_0| = \kappa|A|$. So $|x|$ is dominated by the pulse response of the positive economy with weights $c_i|\ell_i|$. That response has peak $\kappa|A|$ (Lemma S1.1) and total mass $\kappa|A|/(1 - \rho_P)$. Hence $\|x\|_\infty \leq \kappa|A|$ and $\|x\|_2 \leq \|x\|_1 \leq \kappa|A|/(1 - \rho_P)$, and the same holds for the memory states, with an extra factor $1/(1 - r_i)$. *Difference.* Write $c_i \tanh(\ell_i m_{i,t}) = c_i \ell_i m_{i,t} + R_{i,t}$ with $|R_{i,t}| \leq c_i |\ell_i|^3 |m_{i,t}|^3/3$. The linear parts give $(k * x)_t$ with the common kernel. So $e = x^P - x^{P'}$ solves the linear closed loop driven by $u_t = \sum_i R_{i,t} - \sum_j R'_{j,t}$ in place of the probe, and $e = h * u$. By Young's inequality, $\|e\|_2 \leq \|h\|_1 \|u\|_2$, and $\|h\|_1 \leq \kappa/(1 - S_{\mathrm{abs}}) \leq \kappa/(1 - \hat{S})$ as in Theorem 7.2(iii). Also $\| \, |m_i|^3 \|_2 \leq \|m_i\|_\infty^2 \|m_i\|_2 \leq \kappa^3 |A|^3/((1 - r_i)^3 (1 - \rho_P))$, so $\| \sum_i R_i \|_2 \leq \Lambda_P \kappa^3 |A|^3/(1 - \rho_P)$. Combining gives the bound on $\|e\|_2$. For N experiments, the Kullback–Leibler divergence is $N\|e\|_2^2/(2\sigma^2)$. Pinsker's inequality and $\mathrm{TV} \geq 1 - 2\alpha$ give the requirement. (ii) Both markets have $x_0 = \kappa A$ and $m_{i,1} = x_0$, so

$x_1^P - x_1^{P'} = \kappa(\sum_i c_i \tanh(\ell_i \kappa A) - \sum_j c'_j \tanh(\ell'_j \kappa A))$. The linear terms cancel because k_0 is common, and $\tanh u = u - u^3/3 + O(u^5)$ gives the expansion. For $|A|$ small, $|x_1^P - x_1^{P'}| \geq \kappa^4 |\Delta M_3||A|^3/6$. The test that compares the average of N first taps with the midpoint has error $\Phi(-\sqrt{N}\,|x_1^P - x_1^{P'}|/(2\sigma))$, which gives the stated N. □

Theorem 7.4.

Proof. Step 1: probes steer the price. Write F_t for aggregate feedback flow, a function of $x_0, \ldots, x_{t-1}$. Since $x_t = a x_{t-1} + \kappa(F_t + v_t)$ with $\kappa > 0$, the probe $v_t = (x_t^* - a x_{t-1}^*)/\kappa - F_t(x_{<t}^*)$ realizes any target path x^*, and $F_t = (x_t - a x_{t-1})/\kappa - v_t$ is read from the response. So two populations give identical responses to all probes if and only if their flow maps agree on all paths. Because $|F_t| \leq \sum_i c_i |\ell_i||m_{i,t}|$, small target paths need only small probes, which gives the version of part (i).

Step 2: sufficiency of $\tilde{\mu}$. Both shapes are odd, so $c\,\psi(\ell m) = c\,\mathrm{sgn}(\ell)\,\psi(|\ell| m)$ and $F_t = \int \psi(\lambda\, m_t(r))\,\tilde{\mu}(dr, d\lambda)$. Equal $\tilde{\mu}$ give equal flow maps, hence identical responses.

Step 3: identification. Let $D = \tilde{\mu}_P - \tilde{\mu}_{P'} = \sum_k d_k \delta_{(r_k, \lambda_k)}$ with distinct atoms and $\lambda_k > 0$. Take the path with $x_0 = \theta u$ and $x_1 = u$. Then $m_2(r) = r\theta u + u$, and equal flow maps at date 2 give $g(u) = \sum_k d_k \psi(s_k u) = 0$ with $s_k = \lambda_k(r_k\theta + 1)$. This holds for all u (hard limit) or for all u near 0 (tanh, part (i)). For two distinct atoms, $\lambda_k(r_k\theta + 1) = \lambda_j(r_j\theta + 1)$ is an affine equation in θ that is not an identity, so it has at most one root. Fix $\theta > 0$ outside these finitely many roots; then the $s_k > 0$ are distinct.

tanh. g is meromorphic on $\mathbb{C}$ and vanishes near 0, so it vanishes identically and has zero residue at every point. Suppose some $d_k \neq 0$, and let S be the largest s_k with $d_k \neq 0$. At $u^* = i\pi/(2S)$, the term $\tanh(Su)$ has a simple pole with residue $1/S$. For $s_k < S$, $s_k u^* = i\pi s_k/(2S)$ with $0 < s_k/(2S) < \frac{1}{2}$, which is not a pole of tanh. So the residue of g at u^* is $d_k/S \neq 0$, a contradiction. Hence $D = 0$.

A second route for tanh. The odd Taylor coefficients of tanh are $\tau_{2n-1} = 2^{2n}(2^{2n} - 1)B_{2n}/(2n)!$, which are nonzero for every $n \geq 1$ because the Bernoulli numbers B_{2n} are nonzero. If g vanishes near 0, every Taylor coefficient of g vanishes: $\tau_{2n-1} \sum_k d_k s_k^{2n-1} = 0$, so $\sum_k (d_k s_k)(s_k^2)^{n-1} = 0$ for $n = 1, \ldots, K$. This is a Vandermonde system in the distinct positive numbers s_k^2, so $d_k s_k = 0$ and $d_k = 0$ for every k.

Hard limit. g is piecewise linear on $u \geq 0$. At $u = 1/s_k$ only the k-th term changes slope, by $-d_k s_k$, because the s_k are distinct. $g \equiv 0$ forces every slope change to vanish, so $D = 0$. The slope change at $1/s_k$ is visible only on paths with $u \geq 1/s_k$: the state $m_2(r_k)$ must reach the band edge of the traders at (r_k, λ_k). This proves part (ii). □

Proposition 7.5.

Proof. (i) By definition of the stopping time $\tau = \inf\{t \geq 0 : \exists i \text{ s.t. } |\ell_i m_{i,t}| \geq 1\}$, for all $t < \tau$ every trader remains strictly within its linear operating band. Since each trader's policy coincides with linear feedback $w_i m_{i,t}$ on $[0, \tau)$, the aggregate feedback order flow satisfies $F_t(P) = \sum_i w_i m_{i,t}(P)$ on $t < \tau$. By hypothesis, P and P' share the identical feedback kernel k_m. An inductive argument identical to Theorem 7.2(i) establishes that prices, states, and order flows coincide pathwise on $\{t < \tau\}$ across both economies for any causal dealer rule, establishing (i) and the total-variation bound $\mathrm{TV} \leq \Pr(\tau < T)$.

(ii) At breach date $t = \tau$, breached traders in set $\mathcal{B}_t = \{i : |\ell_i m_{i,t}| \geq 1\}$ execute forced liquidations $\Delta y_{i,t} = -\lambda_i(|\ell_i m_{i,t}| - 1)\mathrm{sgn}(\ell_i)$. Under linear causal market clearing, the incremental price impact is $\Delta x_t = \kappa \sum_{i \in \mathcal{B}_t} \Delta y_{i,t} = -\kappa \sum_{i \in \mathcal{B}_t} \lambda_i(|\ell_i m_{i,t}| - 1)$. At step $t+1$, this price shock feeds back into memory updates: $\Delta m_{i,t+1} = \eta_i \Delta x_t + r_i \Delta m_{i,t}$. For a persistent shock with step response

$\sum_{s\geq 0} r_i^s = 1/(1-r_i)$, the cumulative displacement in agent i's decision variable is $\frac{\eta_i}{1-r_i}|\Delta x_t|$. Multiplying by agent i's leverage $|\ell_i|$ and capacity λ_i, the total induced feedback liquidation flow from secondary agents is amplified by the loop gain $\mathcal{G} = \kappa \sum_{i\in\mathcal{B}_t} \lambda_i|\ell_i|\frac{\eta_i}{1-r_i}$. When $\mathcal{G} > 1$, the induced liquidation order flow exceeds the initial stabilizing capacity, driving a monotone self-reinforcing fire-sale cascade with $\mathcal{B}_{t+1} \supset \mathcal{B}_t$. Conversely, when all agents remain within their calm-market envelope ($\vartheta < 1$), the initial breach probability is bounded by Gaussian tail decay $(n+n')e^{-1/(2\vartheta^2)}$ (Theorem 7.2(iii)), bounding the probability of entering the unstable contagion branch. □

Proposition 7.6.

Proof. (i) The proofs of these results use the dealer's rule only through causality and through its being the same map in both markets. (ii) The linear parts of the two expected flows agree because the kernels agree, so the merged weight at each memory is the same. For each trader, $|\mathbb{E}[c\,\mathrm{clip}(Y)] - c\,\mathbb{E}Y| \leq c\,\mathbb{E}[(|Y|-1)_+]$ with $Y = \ell m^+ \sim N(\ell\bar{m}, \sigma^2)$. Since $|Y| \leq |\ell\bar{m}| + \sigma|Z|$, this is at most $c\,\mathbb{E}[(\sigma|Z| - \delta)_+] = c(2\sigma\phi(\delta/\sigma) - 2\delta(1-\Phi(\delta/\sigma))) \leq c\sqrt{2/\pi}\,\sigma e^{-\delta^2/(2\sigma^2)}$. □

Numerical checks. The scripts are `research_extension/nonlinear_scoping/identification_and_figure.py`, `sim_wiener.py`, `sim_passive_clip.py` and `dealer_anticipation.py`. Two populations with the same $\tilde{\mu}$ (a trend follower of capacity 2 and a contrarian of capacity 1, against one trend follower of capacity 1, all at $r = 0.7$ and $|\ell| = 0.1$) give identical responses to 50 random probe sequences of standard deviation 10, with gap exactly 0, under both shapes. The constructive recovery in the main text locates the kinks at 3.702 and 14.813 (true 3.704 and 14.815) and returns net capacities 0.24989 and 0.99988.

S7 Machine checks in detail

Theorem~4.1. `Paper1InventoryEquilibrium.lean` checks the algebraic core of part (i):

- the dealers' first-order condition holds in every state;
- the next price loads on the next flow shock with coefficient κ/D;
- the no-feedback existence condition.

Part (iii) is proved on paper. For part (iv), `Paper1InventoryIFT.lean` applies Mathlib's inverse function theorem along a one-parameter family of feedback $G = \epsilon g$, with g strictly differentiable at $D = 1$. For all small ϵ it gives a zero near $(\kappa_-, 1)$, its uniqueness in a fixed neighbourhood, and the two first-order derivatives in the statement (`p1_inventory_small_feedback`). The version that is uniform over all small cohort measures, in the norm of the statement, is proved on paper.

Theorem 5.1. Lean proves the bound for every K and every horizon T, with a constant that does not depend on T. It uses the theorem's stability hypothesis $|a| + |\kappa|W/(1-r_c-\Delta\eta) \leq \varsigma < 1$, and holds for all error probabilities with $\alpha_0 + \beta_0 \leq 1$ (`p1_thm_kcohort_lower_bound_uniform`): $4\sigma^2(1-\alpha_0-\beta_0)^2 \leq N\,C_U^2(\Delta\eta)^{4K-2}$. The checked pair is a different moment-matched pair with the same order: binomial weights on $2K$ equally spaced nodes, split into odd and even halves. For the Gauss and Gauss–Radau rules themselves, Lean checks the case $K = 3$ (positive weights, matched moments and the fifth-moment gap). Lean also checks both estimator statements for this pair, for every horizon (`p1_thm_kcohort_lower_bound_weights`, `p1_thm_kcohort_lower_bound_W1`). The first covers an estimator of the weights, listed in order of memory rate, that is accurate to

$LW/4^L$ in each coordinate with probability at least 3/4 under both economies. The second covers an estimator of the merged measure that integrates every 1-Lipschitz function vanishing at zero to within $2\Delta\eta W/(\pi(2K-1))$. For measures of equal mass, a Wasserstein error below that accuracy implies this. Both force $\sigma^2 \le N\, C_U^2 (\Delta\eta)^{4K-2}$, here with $L = K-1$. The matching lower order $\|h^A - h^B\|_2 \gtrsim (\Delta\eta)^{2K-1}$ is proved on paper.

Theorem 5.2. Lean verifies this theorem for every $K \ge 1$, using only Lean's standard axioms. The formal proof follows the one above: a Lipschitz bound for the closed-loop deconvolution that does not depend on $\Delta\eta$, the $(\Delta\eta)^{-m}$ rescaling, a sub-Gaussian tail bound over the $2K$ taps, injectivity of the Jacobian of the moment map (proved by Hermite interpolation, without the determinant formula), and a uniform inverse function theorem on the compact class. The formal estimator is the representation with increasing nodes and nonzero weights that matches the $2K$ estimated moments; Lean shows that it is unique and that Prony's linear system has a unique solution whose polynomial has exactly its nodes as roots, so the roots are real. The formal statement allows sub-Gaussian noise and assumes $|r_c| + \Delta\eta \le 1$, so that every memory rate has modulus at most one.

Theorem 5.3. For every K, Lean checks part (i) and part (ii) starting from equal spectra at every frequency (`p1_thm_minphase_zeros`, `p1_thm_minphase_spectrum_H`, `p1_thm_minphase_spectrum_merged`). The check uses H in the rational form $\kappa z D(z)/((z-a)D(z) - \kappa N(z))$ of Theorem 3.3, and *stable* in the sense that the poles of H lie in the open unit disk. Both are derived from the market recursion. The eigenvalues of the transition of the state $(x_t, y_{1,t}, \dots, y_{K,t})$ are exactly the zeros of the denominator of H, so Schur stability of the recursion is the pole condition (`p1_eigen_iff_bottom_root`, `p1_schur_stable_iff_poles_in_disk`). The generating function of the recursion's price response h_t to a unit trade satisfies $\left(\sum_t h_t x^t\right) x^{K+1} B(1/x) = x^{K+1} \kappa x^{-1} D(1/x)$ as formal power series, where B is the denominator of H; with $x = 1/z$ this is $\sum_t h_t z^{-t} = H(z)$ (`p1_impulse_generating`, `p1_tfBottomRev_eval`, `p1_tfTopRev_eval`).

Theorem 5.4. Lean checks the finite-sample consequence of parts (i) and (iii), not the divergence-rate statements. In the checked record the market starts from rest and runs for any number M of warm-up periods before the T recorded prices. The bound does not depend on M, but the passage to the stationary law ($M \to \infty$) is not checked. Part (iii) is checked for every pair of populations with nonnegative merged weights of total at most W, normalized rates in $[-1, 1]$ and equal moments below order $2K-1$ (`p1_thm_passive_cost_matched`). This covers the K-cohort Gauss and Gauss–Radau economies of Theorem 5.1: both rules are positive with nodes in $[-1, 1]$ and integrate polynomials of degree $2K-2$ exactly. Lean proves these properties of the two rules for $K = 2$ and $K = 3$ (`p1_thm_passive_cost_gauss_radau_two`, `p1_thm_passive_cost_gauss_radau_three`), so for these K the check covers the paper's economies with no extra hypothesis. For general K, the exactness of the two quadrature rules, a classical property, enters the check as the moment hypothesis. It also covers the binomial pair used in the numerical checks (`p1_thm_passive_cost_every_K`).

S8 Machine-checked declarations

The table lists, for each main result, the Lean 4 declarations that check it and what is proved only on paper. The project builds against a pinned Mathlib across 123 modules comprising 1,121 named mathematical declarations. For this version the axioms of the headline declarations were verified again via automated kernel extraction: each depends strictly and exclusively on Lean's

standard foundational axioms (`propext`, `Classical.choice` and `Quot.sound`) with zero `sorry` placeholders. A machine check confirms the formal statement, not that it matches the prose. The prose statements above say where the two differ.

Result	Declarations (module)	On paper only
Lemma 2.1	`tracking_gain_explicit_root`, `tracking_prior_riccati_equation` (`Microfoundations.lean`); `p1_optimizing_feedback_euler` (`Paper1StressIdentification.lean`)	Convergence of sample covariances; infinite-horizon value function bounds
Theorem 3.1 (sufficiency)	`p1_eqAggFlow_kernel`, `p1_eqAggFlow_of_kernel`, `p1_kernel_of_eqAggFlow`, `p1_eq_market_invariance`, `p1_eq_insider_profit_invariance`, `p1_eq_group_level_differs` (`Paper1EquilibriumInvariance.lean`)	The unbounded-horizon moment statement; the passage from pathwise equality to equal equilibria
Proposition 3.2	`p1_anylinear_flow_kernel`, `p1_anylinear_iff`, `p1_anylinear_invariance` (`Paper1CoreExtensions.lean`)	The passage from pathwise equality to equal equilibria
Theorem 4.1	`p1_inventory_foc`, `p1_inventory_innovation`, `p1_inventory_benchmark` (`Paper1InventoryEquilibrium.lean`); `p1_inventory_small_feedback` (`Paper1InventoryIFT.lean`)	Part (iii); the version of part (iv) uniform over small cohort measures
Theorem 4.2	`p1_eq_realize_D`, `p1_eq_realize_dealer`, `p1_eq_scaled_cohort` (`Paper1EquilibriumReducedForm.lean`)	Parts (ii)–(iii), and part (iv) beyond the realization
Theorem 5.1	`p1_thm_kcohort_lower_bound_uniform` (`Paper1KCohortUniform.lean`); `p1_thm_kcohort_lower_bound_weights`, `p1_thm_kcohort_lower_bound_W1` (`Paper1KCohortEstimators.lean`); `p1_passive_fisher_le_active` (`Paper1StressIdentification.lean`)	The matching lower order of the response gap; the checked pair is a binomial moment-matched pair of the same order
Theorem 5.2	`p1_thm_prony_upper_bound` (`Paper1PronyUpper.lean`)	—

Result	Declarations (module)	On paper only
Theorem 5.3	`p1_thm_minphase_zeros` (`Paper1MinPhase.lean`); `p1_thm_minphase_spectrum_H`, `p1_thm_minphase_spectrum_merged` (`Paper1SpectralH.lean`); `p1_closed_loop_transmission_zero` (`Paper1StressIdentification.lean`); model link `p1_eigen_iff_bottom_root`, `p1_schur_stable_iff_poles_in_disk`, `p1_impulse_generating` (`Paper1ModelLink.lean`)	—
Theorem 5.4	`p1_thm_passive_cost_matched`, `p1_thm_passive_cost_gauss_radau_two` (`Paper1PassiveMatched.lean`); `p1_thm_passive_cost_gauss_radau_three` (`Paper1GaussRadau3.lean`); `p1_thm_passive_cost_every_K` (`Paper1PassiveKCohort.lean`); `p1_higher_moment_sample_size_lower` (`Paper1StressIdentification.lean`)	The divergence-rate statements; the stationary limit; Gauss–Radau exactness for general K enters as a hypothesis
Theorem S2.3	`p1_insider_reduced_form`, `p1_insider_best_response`, `p1_insider_kyle_limit`, `p1_insider_posterior_variance` (`Paper1EquilibriumInsider.lean`)	The fixed-point characterization in part (i), part (iii), and the numerical comparative statics
Proposition 6.1	`p1_prop_target_two_cohorts`, `p1_prop_target_every_K` (`Paper1SupervisorySeparation.lean`)	The checked two-group pair is a different moment-matched pair from the Gauss and Gauss–Radau pair, with the same orders; the every-K version assumes a dominant-group margin; for the quadrature pairs the largest-weight comparison is checked numerically for $K \leq 200$, not proved
Proposition 6.2	`p1_tagged_iv_determinant_bound` (`Paper1TaggedIV.lean`); `p1_iv_boundary_decomposition` (`Paper1IVBoundary.lean`)	The stationary IV central limit theorem
Theorem 7.1	`p1_sat_sufficiency`, `p1_flow_market_invariance` (`Paper1SaturatingSufficiency.lean`)	The passage from pathwise equality to equal equilibria

Result	Declarations (module)	On paper only
Theorem 7.2	`p1_sat_stress_invariance`, `p1_sat_follows_linear`, `p1_satAggFlow_linear_of_band`, `p1_sat_capacity_differs` (`Paper1SaturatingSufficiency.lean`)	Parts (ii)–(iv): the total-variation, Gaussian and probe bounds
Proposition 7.3	`p1_cubic_pulse_tap1_kernel_cancel`, `p1_cubic_pulse_tap1_price_gap` (`Paper1StressIdentification.lean`)	Part (i) and continuous tanh bounds; part (ii) linear cancellation and A^3 separation are machine-checked
Theorem 7.4	`p1_probe_steer_target`, `p1_probe_flow_recovery`, `p1_probe_flow_recovery_eq`, `p1_odd_policy_decomp`, `p1_sat_signed_capacity_sufficiency`, `p1_vandermonde_2x2_invertible`, `p1_vandermonde_2x2_weights_zero`, `p1_hard_limit_kink_recovery` (`Paper1StressIdentification.lean`)	General meromorphic residue route; Step 1, Step 2, Step 3 2×2 Vandermonde separation and hard-limit kink slope recovery are machine-checked
Proposition 7.5	`p1_delever_stress_invariance` (`Paper1CoreExtensions.lean`); `p1_firesale_loop_gain_cascade`, `p1_firesale_calm_contract` (`Paper1StressIdentification.lean`)	Continuous Gaussian tail bounds; part (i) path invariance and part (ii) fire-sale loop gain cascade are machine-checked
Proposition 7.6	`p1_dealer_expected_linear_invariance`, `p1_clip_in_band` (`Paper1StressIdentification.lean`)	Continuous normal expectation integral; part (i) expected linear invariance and in-band truncation identity are machine-checked

S9 Notation

Symbol	Meaning
x_t	pricing error (price minus fundamental)
a	price persistence (anchor), $0 \le a < 1$
κ	price impact per unit of executed quantity
$q_i,\ \eta_i,\ r_i$	cohort weight, adaptation rate, memory rate $r_i = 1 - \eta_i$
$w_i = q_i \eta_i$	merged weight
$M = \sum_i w_i \delta_{r_i}$	merged rate–weight measure
$k_m,\ G(z)$	feedback kernel and its transform
$H(z),\ h_j$	closed-loop transfer function and impulse response, $h_0 = \kappa$
$J,\ S$	state transition matrix and $S = a + \kappa \sum q_i$
$v_t,\ V$	executed probe quantity and total quantity $\sum \lvert v_t \rvert$

Symbol	Meaning
d_T	impulse-response difference between hypotheses, first T taps
σ	observation-noise standard deviation
$\delta,\ \Delta\eta$	half-separation of a known pair; spacing of unknown cohorts
K_*	transfer-difference constant $2\kappa^2 q/[\epsilon^3(1-S)^2]$
T_h	Toeplitz matrix $(h_{\lvert i-j\rvert})$ of execution cost
$g_i(\omega)$	dissipation kernel $\mathrm{Re}\,\hat{H}_i(e^{i\omega}) - \kappa/2$
$B,\ \mu,\ c,\ \phi$	capital, maintenance-margin fraction, proportional cost, taker fee
V_{eff}	per-trade execution capacity $\min(V_{\max}, D_{\max}/\kappa)$
$\hat{Q},\ \hat{S}$	reduced-form loop gain and persistence indices